



In-Country Working Session: FASTR Implementation & RMNCAH-N Service Monitoring Analysis

January 27–30, 2026 | Lusaka

STRENGTHENING HEALTH SYSTEMS AND RMNCAH-N OUTCOMES THROUGH RAPID CYCLE ANALYTICS AND DATA USE

*Country Workshop: Introduction to FASTR
RMNCAH-N service Use Monitoring*

**Lusaka, Zambia
27–30 January 2026**



WELCOME AND OPENING REMARKS

TBD

Introductions



Agenda

Time	Agenda	Facilitator/Presenter
Opening Session		
08:30–09:00	Participant registration	MoH team
09:00–09:10	Welcome and opening remarks	MoH team
09:10–09:20	Icebreakers/Introductions	MoH team
09:20–09:35	Overview of agenda, workshop objectives	GFF FASTR team
Session 1: Overview of the FASTR approach		
09:35–10:30	Overview: FASTR Approaches	GFF FASTR team
Session 2: HMIS data extraction		
10:30–11:30	Data extraction: Rationale and methods	GFF FASTR team
Session 3: Introduction to the FASTR analytics platform		
11:30–12:30	Introduction to the FASTR analytics platform	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
14:00–14:30	Getting participants into the platform	GFF FASTR team
Session 4: Configuring the FASTR analytics platform		
14:30–16:30	Configuring the analysis platform	GFF FASTR team

Workshop Objectives

The objectives of the workshops are to:

- Introduce the FASTR HMIS analysis platform
 - Familiarize participants with the platform's purpose, core features, and functionalities, including workflows, reporting options, and outputs.
- Strengthen analytical and reporting skills
 - Enhance participants' capacity to manage data structures, customize visualizations, and produce tailored reports for diverse audiences.
- Prepare MoH for participation in the multi-country workshop
 - Configure the FASTR analytics platform for Zambia
 - Produce the first quarterly report
- Adapt the FASTR phone survey questionnaire to the Zambian context

Your Expectations for the Workshop

Write on a sticky note and stick it on the flip chart:

- What do you want to learn?
- What questions about FASTR do you want answered?

We will review these on day four to see if we met our expectations.



Expected Outputs

Disruption Analysis Activities

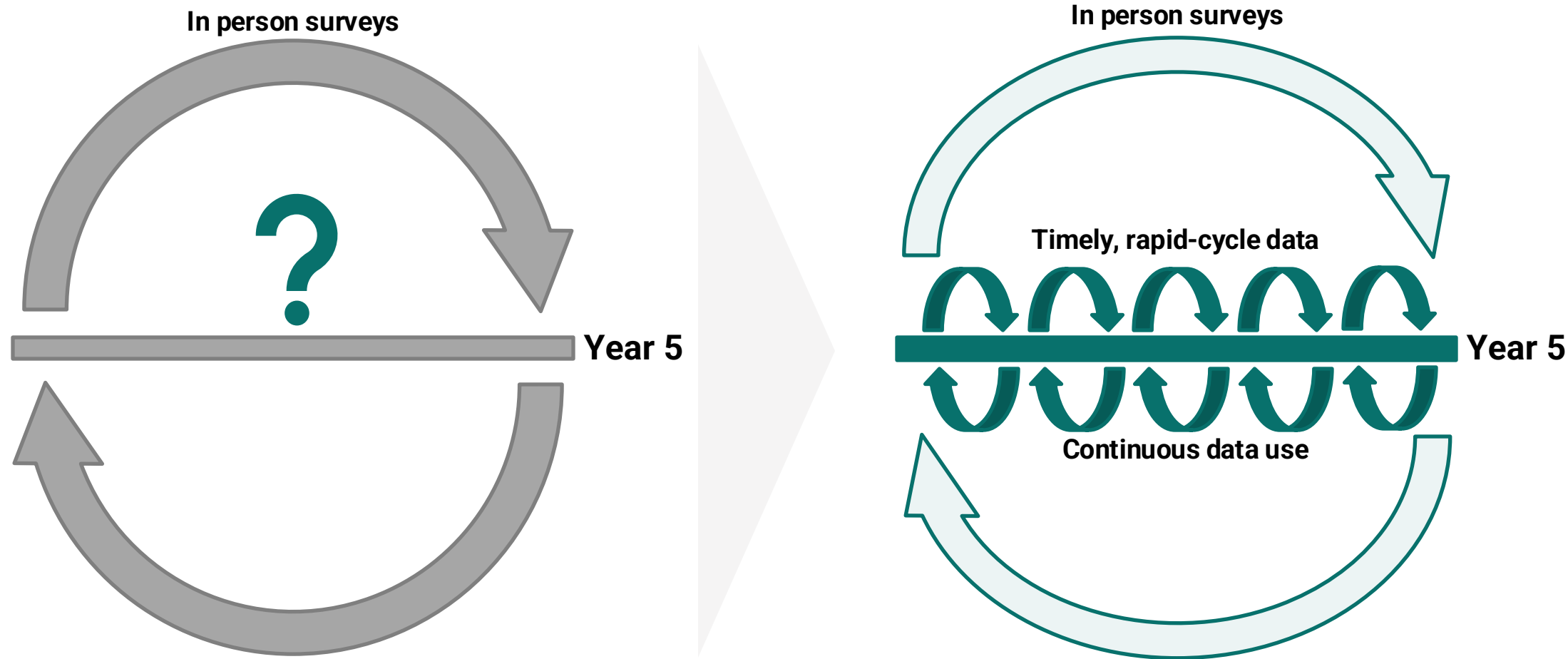
- Data extraction and configuration of analytics platform
- Produce first quarterly report
- Review results and contextualize findings (with relevant program teams)
- Roadmap for participation in Abuja workshop

Phone Survey Activities

- Adaptation of questionnaire (with program input)

Session 1a: Overview of the FASTR approach

What are we trying to achieve? Rapid cycle analytics accelerates improvements in RMNCAH-N outcomes by increasing the systematic use of data for decision making



How can this be achieved? Timely, rigorous, and low-cost approaches to monitoring PHC systems, underpinned by capacity building and data use support aligned with country demand and needs



RMNCAH-N Service Use

Leverage HMIS data to routinely monitor changes in the coverage of health services during reforms, shocks, or within vulnerable populations, while identifying and addressing data quality challenges.



Facility Surveys

Assess gaps in primary health care service delivery; understand the impact of health reforms or shocks on PHC performance.



HH/client surveys

Provide snapshots of health service utilization, foregone care, and community perceptions of service quality



Rapid Qualitative Studies

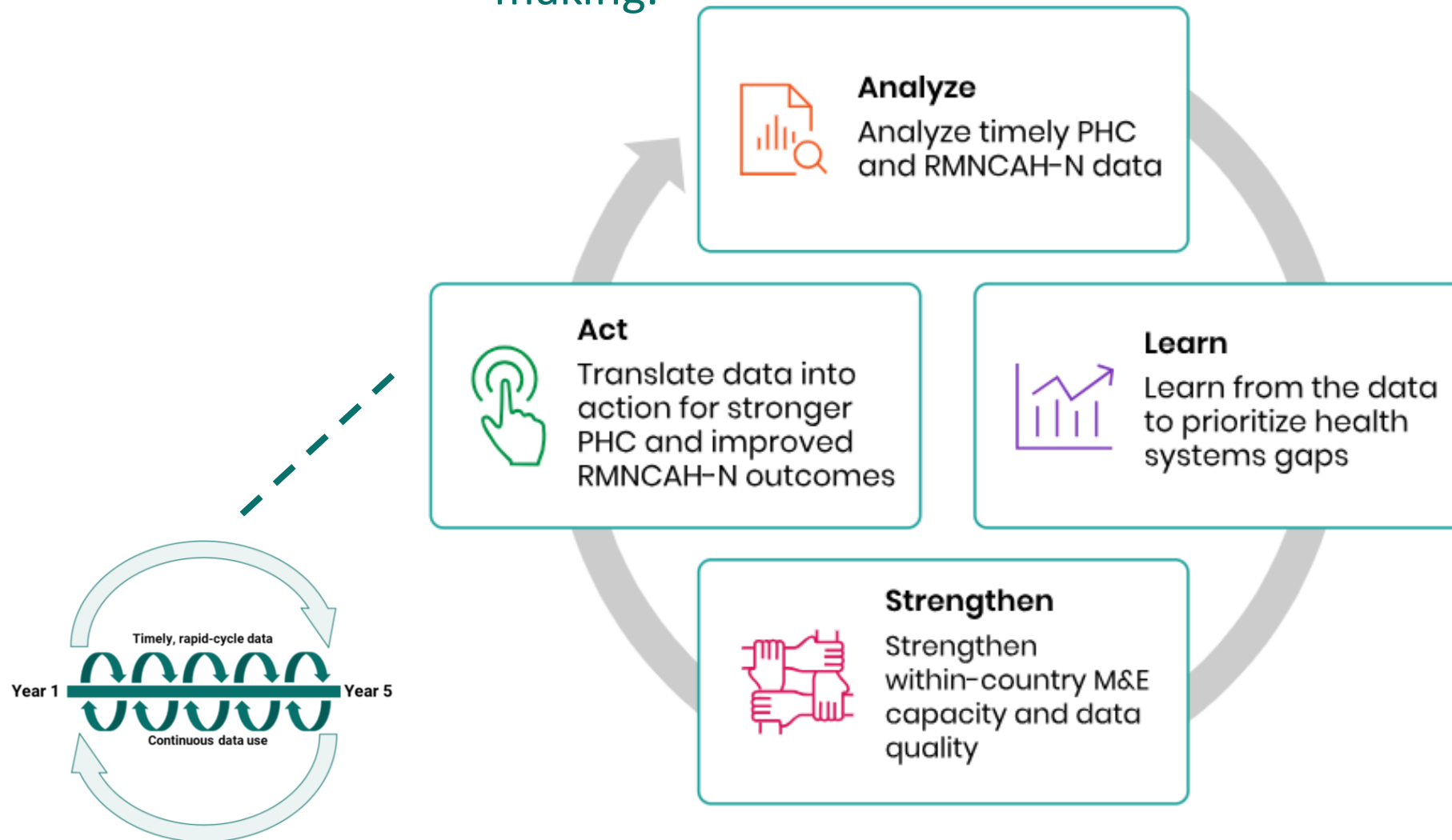
Gain a richer understanding of emerging health systems issues through rapid and adaptable mixed-methods approaches.

Build capacity + data use support

Building within-country capacity for data collection, analysis, and use including data reviews and subnational feedback loops.

What is FASTR?

An **approach** to catalyzing continuous '*analyze, learn, strengthen, act*' cycles to drive the systematic use of timely data for decision making.

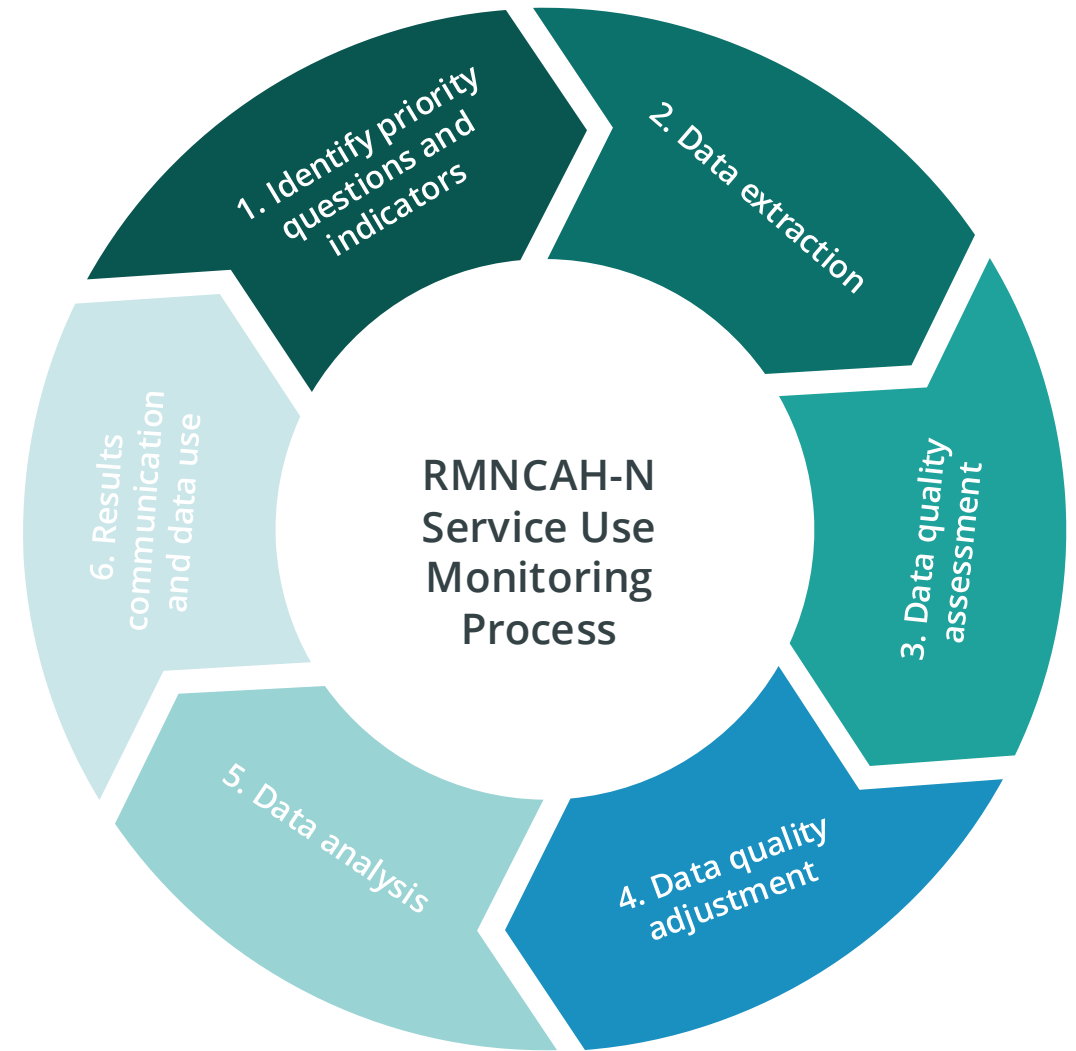


What is the FASTR approach to RMNCAH-N service use monitoring?

Quarterly analyses of DHIS2 data, focusing on prioritized national indicators

Building **sustainable tools** to ensure that stakeholders who need to use data can generate the right analysis and visualizations, at the right time, on their indicators of interest

Combining analysis and visualization with capacity strengthening and data use support for **sustainability and institutionalization**.



Demo – FASTR analysis platform



Session 1b: Identifying priority questions and indicators

What is a data use case?

A data use case is a specific scenario where data is utilized to achieve a particular goal or solve a problem.

Why is defining a data use case important?



Guides decision making by providing a clear framework for analysis

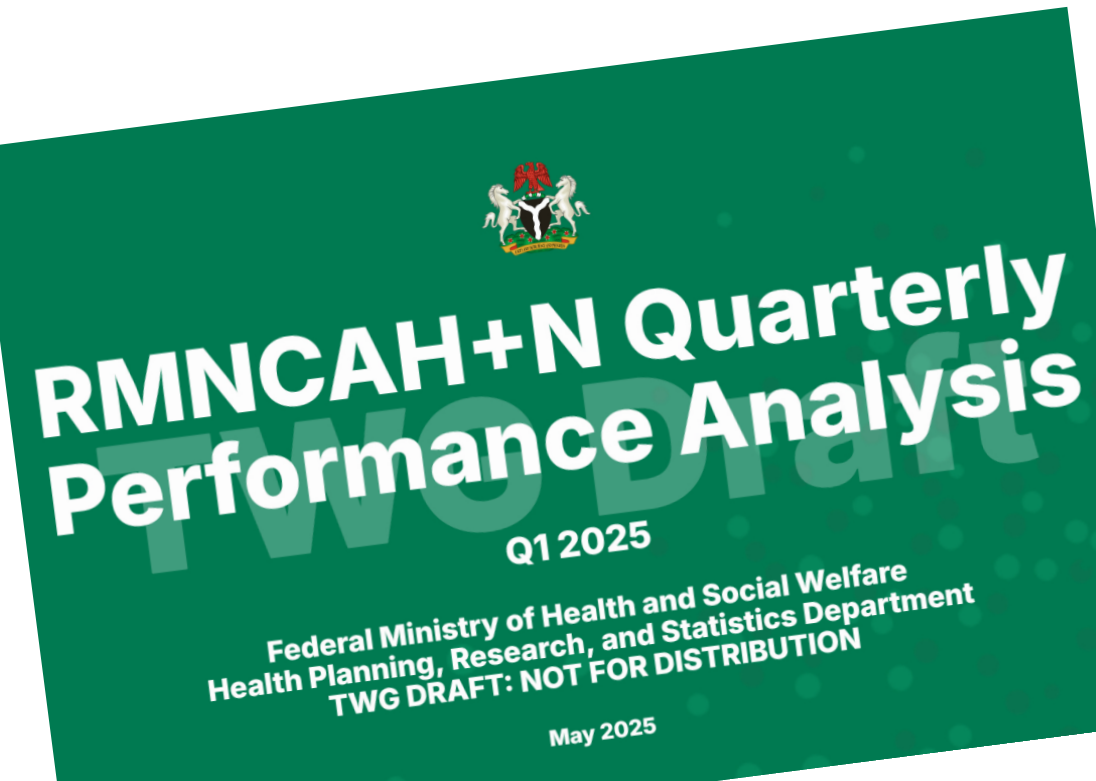


Enhances efficiency by focusing analyses on a set of relevant key indicators to solve a specific data need



Leads to better results by aligning data efforts with organizational goals.

In Nigeria, FASTR is enabling quarterly performance monitoring for the National Health Sector Renewal Initiative – and informing systems-strengthening efforts along the way



1.3 Mean DQA score, by state and year

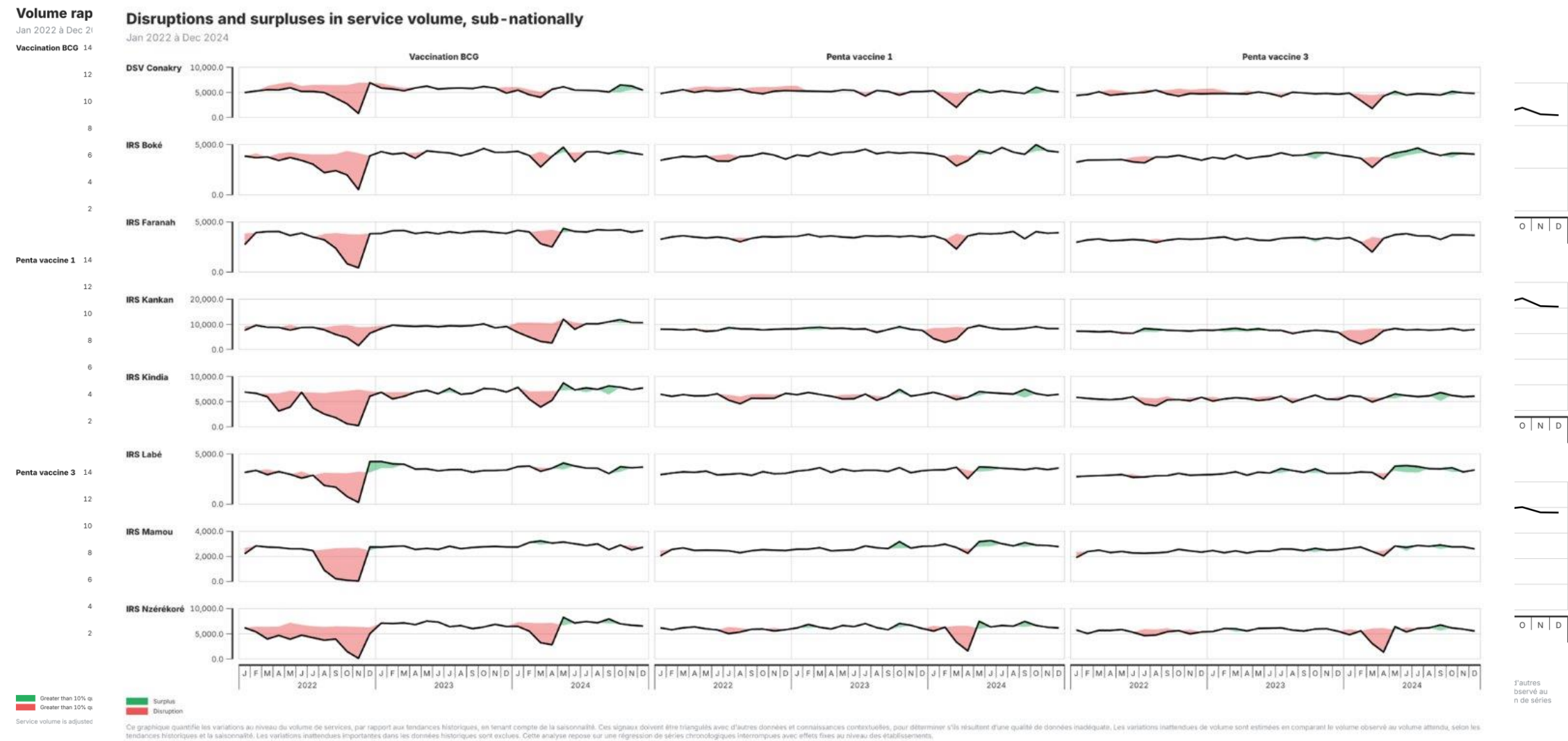
Average data quality score across facility-months from 2020 - 2025

	2020	2021	2022	2023	2024	2025
ab Abia State	76.4%	84.5%	85.7%	82.3%	87.5%	90.3%
ad Adamawa State	88.4%	87.9%	89.0%	90.2%	90.6%	88.2%
ak Akwa-Ibom State	92.5%	90.8%	92.6%	89.4%	93.4%	94.8%
an Anambra state	88.5%	85.8%	85.7%	84.1%	84.1%	84.6%
ba Bauchi State	86.3%	86.4%	86.5%	83.4%	83.3%	84.5%
be Benue State	80.6%	81.2%	82.3%	85.0%	85.1%	83.3%
bo Borno State	86.9%	85.9%	87.7%	90.7%	90.5%	87.4%
by Bayelsa State	78.2%	78.0%	80.6%	82.9%	87.4%	89.3%
cr Cross River State	88.9%	87.5%	88.9%	90.9%	86.6%	92.1%
de Delta State	51.6%	53.4%	42.5%	45.1%	46.2%	47.5%
eb Ebonyi State	88.6%	87.9%	90.3%	88.1%	89.7%	80.9%
ed Edo State	82.3%	79.9%	80.4%	80.3%	83.2%	74.7%
ek Ekiti State	80.7%	78.3%	77.4%	81.6%	83.8%	86.2%
en Enugu State	77.1%	80.4%	78.6%	82.1%	89.7%	86.1%
fc Federal Capital Territory	79.2%	82.3%	79.4%	82.8%	83.8%	72.8%
go Gombe State	87.8%	87.8%	87.8%	84.5%	83.1%	84.9%
im Imo State	88.5%	85.3%	85.7%	84.0%	87.3%	85.2%
ji Jigawa State	92.0%	91.5%	88.8%	90.3%	90.7%	90.1%
kd Kaduna State	91.3%	88.7%	91.0%	90.3%	90.6%	91.4%
ke Kebbi State	81.1%	79.9%	80.4%	78.1%	77.8%	78.1%
kn Kano State	93.2%	93.8%	93.7%	91.9%	88.6%	87.7%
ko Kogi State	71.8%	64.5%	76.8%	81.7%	84.7%	83.0%
kt Katsina State	45.6%	45.0%	44.8%	45.7%	46.9%	47.6%
kw Kwara State	87.0%	86.2%	86.8%	82.8%	82.2%	83.7%
la Lagos State	68.5%	65.3%	62.6%	61.9%	68.2%	69.0%
na Nasarawa State	85.7%	86.5%	87.8%	87.2%	87.2%	85.1%
ni Niger State	86.1%	85.2%	84.3%	85.6%	87.3%	90.8%
og Ogun State	81.5%	78.7%	77.7%	79.8%	84.5%	73.6%
on Ondo State	91.4%	87.8%	84.8%	87.7%	88.1%	88.3%
os Osun State	82.7%	83.3%	83.3%	83.9%	85.3%	86.9%
oy Oyo State	81.7%	82.1%	79.7%	87.1%	86.4%	84.1%
pl Plateau State	80.6%	81.9%	82.0%	81.8%	84.8%	86.8%
ri Rivers State	73.2%	76.1%	78.8%	71.3%	79.3%	57.1%
so Sokoto State	53.5%	43.4%	63.8%	57.6%	62.4%	66.6%
ta Taraba State	45.4%	44.0%	44.8%	43.4%	45.0%	47.4%
yo Yobe State	85.9%	86.4%	84.7%	87.3%	88.2%	89.4%

"Part of the FASTR process should open up serious conversations that will lead to improvements in the system. It's not about pulling an indicator that will not get us good information. It is about using this FASTR process to make better decisions and re-engineer the system"

Dr. Anthony Adoghe, Director of M&E, Federal Ministry of Health and Social Welfare

Quarterly RMNCAH-N service use monitoring has supported Guinea to track investment case progress and quickly spot challenges – like this sharp drop in immunization in the beginning of 2024.

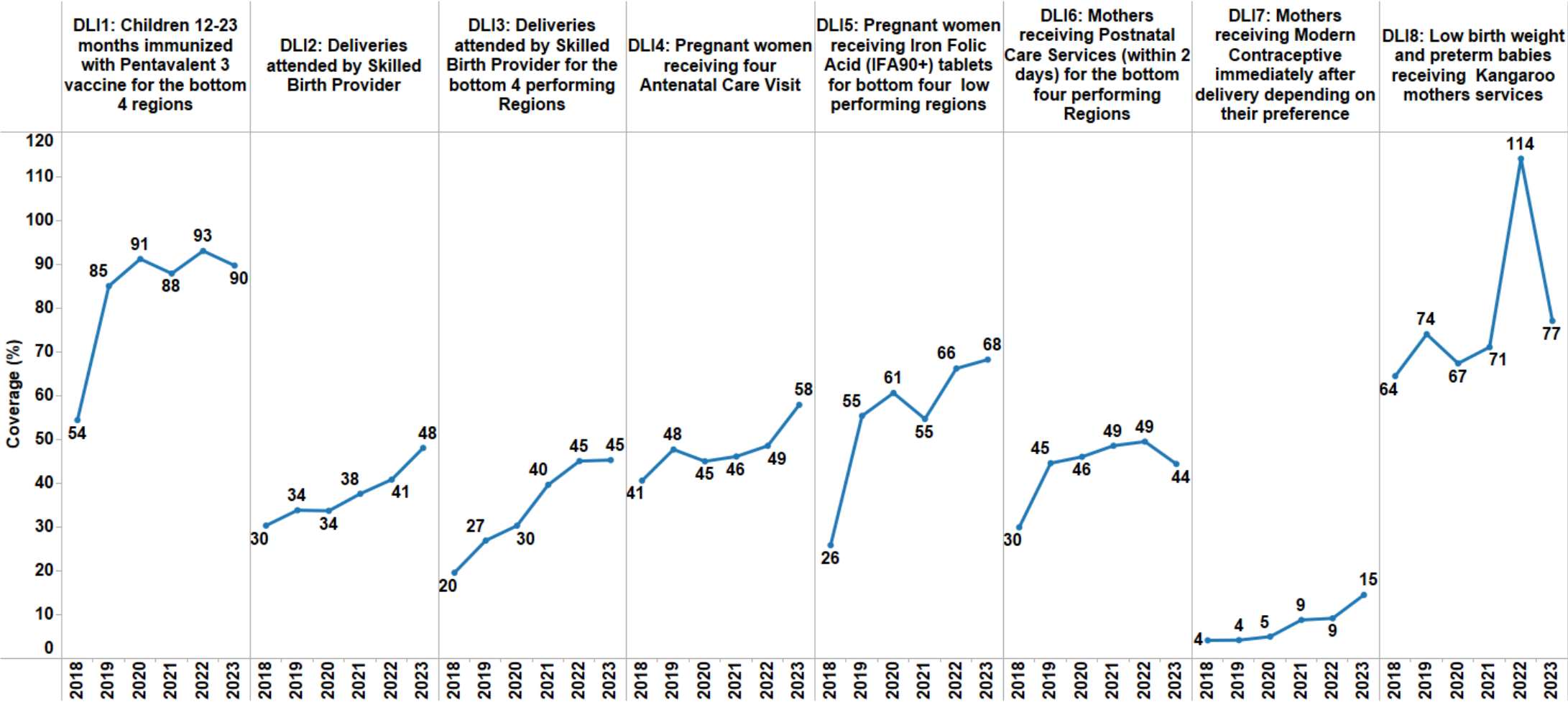


f'autres
bservé au
n de séries

Using FASTR HMIS analysis for monitoring results prioritized in government program supported by WB/GFF in Ethiopia

Project Disbursement Linked Indicators

DHIS2, 2018-2023 [12/2015 Eth. calendar]



*Excluding conflict-affected Woredas, adjusted for completeness and outliers

What is our common data use case that will be the focus of this workshop?

Following large shifts in resource availability from external sources, many countries are experiencing abrupt and dramatic reductions in financing

- ▶ Resulting in critical gaps in programs and systems
- ▶ Leading to potentially severe effects on service delivery and health outcomes for women, children and adolescents

Key questions arising:

- What is the magnitude of the cuts, and what effect are they having on service delivery?
- What is the optimal way to prioritize remaining resources?
- What other adaptations can safeguard and strengthen essential service delivery for women, children and adolescents?

How do we select indicators? What makes a good indicator for the FASTR analysis?

Indicator selection is critical to the quality and usefulness of FASTR analysis. Indicators should be chosen based on the following criteria:

- **Relevance**
Does this indicator provide data that aligns with our priority questions and objectives?
- **Volume**
Is this indicator collected at a high volume, which improves the robustness of analysis?
- **Completeness**
Does the indicator have a high completeness rate across reporting facilities?
- **Frequency**
Is the indicator reported frequently enough (e.g., monthly) to support rapid-cycle analysis?
- **Type**
Is this indicator a count of services delivered?

FASTR core indicators

The FASTR approach focuses on a core set of RMNCAH-N indicators that represent key points along the reproductive, maternal, newborn, child, and adolescent health and nutrition continuum in low- and middle-income countries.

These indicators typically have higher reporting volumes and completeness and serve as proxies for broader service delivery patterns.

- | | |
|---|-------------------------------------|
| 1. Antenatal client 1 st visit | 5. BCG doses |
| 2. Antenatal client 4 th visit | 6. Pentavalent 1 st dose |
| 3. Institutional delivery | 7. Pentavalent 3 rd dose |
| 4. Postnatal care 1 | 8. Outpatient visits |

Why focus on high-volume indicators?

One of the core strengths of the FASTR approach is its ability to adjust for data quality issues. High-volume indicators are better suited to this process because:

- **Reduced sensitivity to outliers**
In low-volume indicators, individual data points can disproportionately affect trends.
- **More stable estimates**
High-volume data reduce random variability and improve the reliability of trend detection.
- **Clearer identification of true anomalies**
Larger counts make it easier to distinguish genuine outliers from natural variation.

Why focus on high-completeness indicators?

Indicators with high reporting completeness are preferred because they:

- **Improve data reliability**
More complete data reduces bias and provide a more representative picture of service delivery.
- **Support consistent analysis**
High completeness enables meaningful comparisons across time and geographic areas.
- **Reduce misinterpretation**
Incomplete data can falsely suggest changes in service utilization when changes are driven by reporting gaps rather than real trends.
- While statistical methods such as imputation can be used to address incomplete data, these methods require assumptions about missing values.

Why focus on count indicators?

- **Limitations of proportion indicators**
 - Proportions limit the ability to adjust numerators and denominators separately for data quality issues.
 - Numerators and denominators may each be affected by different sources of error.
 - Separating counts from denominator estimation allows for more transparent and flexible adjustment.
- **Mortality as a rare event**
 - Mortality indicators are typically low-frequency and not well suited to adjustment.
 - These indicators are generally better analyzed using annual rather than monthly or quarterly data.

Countries have selected indicators to align with the disruptions context and country priorities

- The FASTR Data Prep Checklist has been shared with countries.

		Indicator mapping for RMNCAH-N Service Use Monitoring using Administrative Data				
Data Source Mapping				Indicator Definition		
INDICATOR CATEGORY	INDICATOR OF INTEREST	INDICATOR SOURCE (national strategy, WB project, FASTR core indicators, etc)	IS THIS INDICATOR IN DHIS2?	OFFICIAL INDICATOR NAME IN DHIS2	INDICATOR JSON ID	NOTES
RMNCAH+N	ANC1	FASTR core indicators				
RMNCAH+N	ANC4	FASTR core indicators				
RMNCAH+N	Deliveries - institutional	FASTR core indicators				
RMNCAH+N	BCG	FASTR core indicators				
RMNCAH+N	PNC1	FASTR core indicators				
RMNCAH+N	Penta1	FASTR core indicators				
RMNCAH+N	Penta3	FASTR core indicators				
RMNCAH+N	OPD visits	FASTR core indicators				

- The checklist includes the FASTR core RMNCAH-N indicators
- Countries have added additional indicators that may be particularly relevant to the country context (e.g., indicators related to services expected to be impacted by recent funding shifts, priority indicators for the government and/or the WB project).
- These are the indicators included in the current analysis. Countries can continue to add indicators over time as needed for their use case(s).

Activity

We are going to review the Zambia data prep checklist together and discuss for each indicator:

- **Relevance**
Does this indicator provide data that aligns with our priority questions and objectives?
- **Volume**
Is this indicator collected at a high volume, which improves the robustness of analysis?
- **Completeness**
Does the indicator have a high completeness rate across reporting facilities?
- **Frequency**
Is the indicator reported frequently enough (e.g., monthly) to support rapid-cycle analysis?
- **Type**
Is this indicator a count of services delivered?

Based on the discussion we can refine the selection of indicators.



Session 2: HMIS data extraction

Show of hands

Do you regularly extract data from DHIS2?

If so, what are the primary reasons?

Why would you extract data from DHIS2? Why not just do analysis in DHIS2 itself?

Data quality adjustment

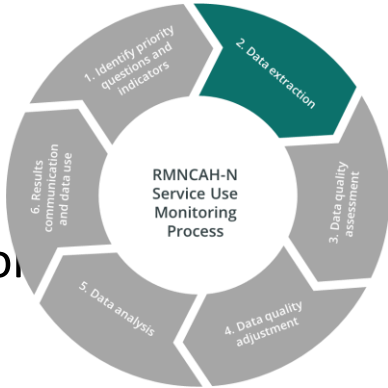
- The FASTR approach focuses on data quality adjustments to expand the analyses countries can do with DHIS2 data and to generate more robust estimates.

Analysis complexity

- The FASTR approach uses more advanced statistical methods, such as regression analysis, which are not available in DHIS2. While DHIS2 can plot trends over time using raw data, FASTR can go further by identifying significant increases or decreases in service volume, adjusting for data quality issues, accounting for expected seasonal variations, and comparing key periods, such as before and after a reform.

The choice between DHIS2 and the FASTR approach should be guided by the specific purpose of your analysis. Select the tool that best aligns with your analytical needs!

Data format and granularity



- Data should be downloaded for each *indicator of interest*, at *facility level*, and *monthly* for the *period of interest*.
- Data should be saved in long format meaning each row represents a single observation or measurement (see example).
- Data should be saved in .csv files which

	A	B	
1	orgunitlevel1	orgunitlevel2	o
2	MOH MALAWI Govt	Central West Zone	Li
3	MOH MALAWI Govt	Central West Zone	Li
4	MOH MALAWI Govt	Central West Zone	Li
5	MOH MALAWI Govt	Central West Zone	Li
6	MOH MALAWI Govt	South East Zone	Ba
7	MOH MALAWI Govt	South East Zone	Ba
8	MOH MALAWI Govt	South East Zone	Ba
9	MOH MALAWI Govt	South West Zone	Ba
10	MOH MALAWI Govt	South West Zone	Ba
11	MOH MALAWI Govt	South West Zone	Ba
12	MOH MALAWI Govt	South West Zone	Ba
13	MOH MALAWI Govt	South West Zone	Ba
14	MOH MALAWI Govt	Central West Zone	N
15	MOH MALAWI Govt	Central West Zone	N
16	MOH MALAWI Govt	Central West Zone	N
17	MOH MALAWI Govt	North Zone	M
18	MOH MALAWI Govt	North Zone	M
19	MOH MALAWI Govt	North Zone	Mzimba-North-DHO
20	MOH MALAWI Govt	Central West Zone	Dedza-DHO
21	MOH MALAWI Govt	Central East Zone	Nkhotakota-DHO
22	MOH MALAWI Govt	Central East Zone	Nkhotakota-DHO
23	MOH MALAWI Govt	South West Zone	Chikwawa-DHO
24	MOH MALAWI Govt	Central West Zone	Ntcheu-DHO
25	MOH MALAWI Govt	South East Zone	Zomba-DHO

Sb8xVmXyL5p	Banja La Mtsogolo Mzuzu	550	202307	Jul-23	202307	DtefmOHJHP9	RHD FP Contraceptive	Method IUCD Insertion	14	8/24/2023
spQpsut9f1B	Bembeke Health Centre	1701	202307	Jul-23	202307	DtefmOHJHP9	RHD FP Contraceptive	Method IUCD Insertion	3	8/24/2023
n6UZ8o9yo6L	Benga Health Centre	1101	202305	May-23	202305	DtefmOHJHP9	RHD FP Contraceptive	Method IUCD Insertion	2	8/24/2023
n6UZ8o9yo6L	Benga Health Centre	1101	202307	Jul-23	202307	DtefmOHJHP9	RHD FP Contraceptive	Method IUCD Insertion	1	8/24/2023
FO0kNPmm5E9	Bereu Dispensary	3322	202307	Jul-23	202307	DtefmOHJHP9	RHD FP Contraceptive	Method IUCD Insertion	1	8/24/2023
pPHM09AVDAS	Bilira Health Centre	1801	202307	Jul-23	202307	DtefmOHJHP9	RHD FP Contraceptive	Method IUCD Insertion	4	8/24/2023
gS8A33Rsqrj	Bimbi Health Centre	2701	202307	Jul-23	202307	DtefmOHJHP9	RHD FP Contraceptive	Method IUCD Insertion	1	8/24/2023



Why monthly facility level data?

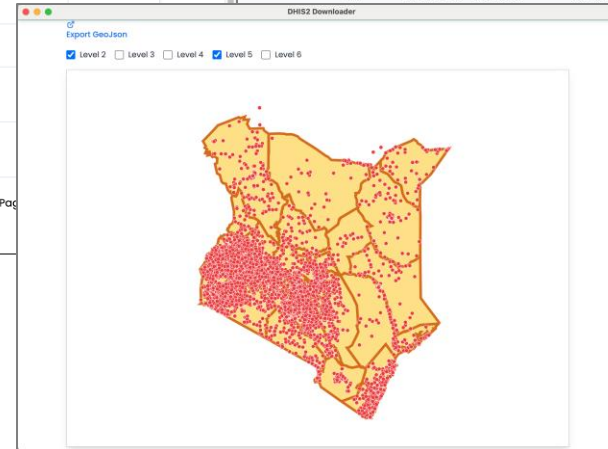
How much data?

Initial FASTR analysis

- Generally recommended to download approximately **five years** of historical data.
- However, the exact period should be determined based on data availability, consistency in indicator definitions over time, and the specifics of a country's routine data system.
- Ideally, using at least five years of historical data allows for a thorough assessment of trends over time.

Routine update to FASTR analysis

- Start with the existing database and download new data covering the most recent months not previously included – this is usually a three-month period when the FASTR analysis is being implemented on a quarterly basis.
- Additionally, include the three preceding months to the new data time period, as this relatively recent data is often subject to changes due to late reporting or data quality adjustments.
- If you have reason to believe there have been substantial changes to the historical data, you can always choose to redownload a longer time period.



Data extraction

The **FASTR** analytics platform contains a direct import feature to automatically import data from DHIS2. This is often the easiest approach once indicators have been identified for inclusion in the platform.

Demonstration site

FASTR

Projects

Data

Assets

Users

Settings

Ashley Sheffield (Admin)

<

IMPORT IN PROGRESS

HMIS Data

1

2

3

4

5

< Prev

> Next

↺

Discard import

☐ Upload CSV file

☒ Import directly from DHIS2

Save

Demonstration site

FASTR

Projects

Data

Assets

Users

Settings

Ashley Sheffield (Admin)

<

IMPORT IN PROGRESS

HMIS Data

1

2

3

4

5

< Prev

> Next

↺

Discard import

DHIS2 Import Configuration

Enter your DHIS2 connection details to import data.

DHIS2 URL

https://example.dhis2.org

DHIS2 Username

username

...

DHIS2 Password

password

...

☐ Save credentials for this session

Confirm and continue

Demonstration site

FASTR

Projects

Data

Assets

Users

Settings

Ashley Sheffield (Admin)

<

IMPORT IN PROGRESS

HMIS Data

1

2

3

4

5

< Prev

> Next

↺

Discard import

DHIS2 Data Selection

Select indicators to import

Select Period Range

INDICATOR ID	LABEL	COMMON IDS
BfzfJfK54Pe	Oncho treatment population treated_nrd	
h1S137QQA0	Vitamin A supplement 6-11 months	vitamin_a
bjnthgrPyk	Client tested for HIV through PICT (F)	
Bktgxn0aSD5	IMAM OTP Discharged - cured	
brJQU0Xe17w	Schisto treatment population treated_nrd	
C173CzesDAC	Client HIV+ screened for TB total	
c7D2k8NmWU4	Injectables (Depo) - Postpartum FP	fp_pp_acceptors
CmcPHccfW	Male Condoms New Client	fp_short, fp_all_users, fp_new_users
Cophiw05plu	Implant (5 years) (Ladelle) New Client	fp_long, fp_all_users, fp_new_users
CUDsau0PJfr	IUDs Postpartum FP	fp_pp_acceptors
cV1HxydyoJe	Client tested HIV positive (M)	
dc50tAPleaB	Client tested HIV positive (F)	
Em44duyT5id	Malaria - Adult tested by microscopy	malaria_tested
exH2Ff3keK	Child screened Weight For Height	with_screened
F5DCeEMZbP	Pentavalent 3rd dose	penta3
F83VoRCwxxX	Pentavalent 1st dose	penta1

Feb 2025 to Jan 2026

Selected: 0 indicators × 12 periods = 0 data points to fetch

Save selection

DHIS2 Data Downloader

The Data Downloader is a desktop application for extracting data from DHIS2.

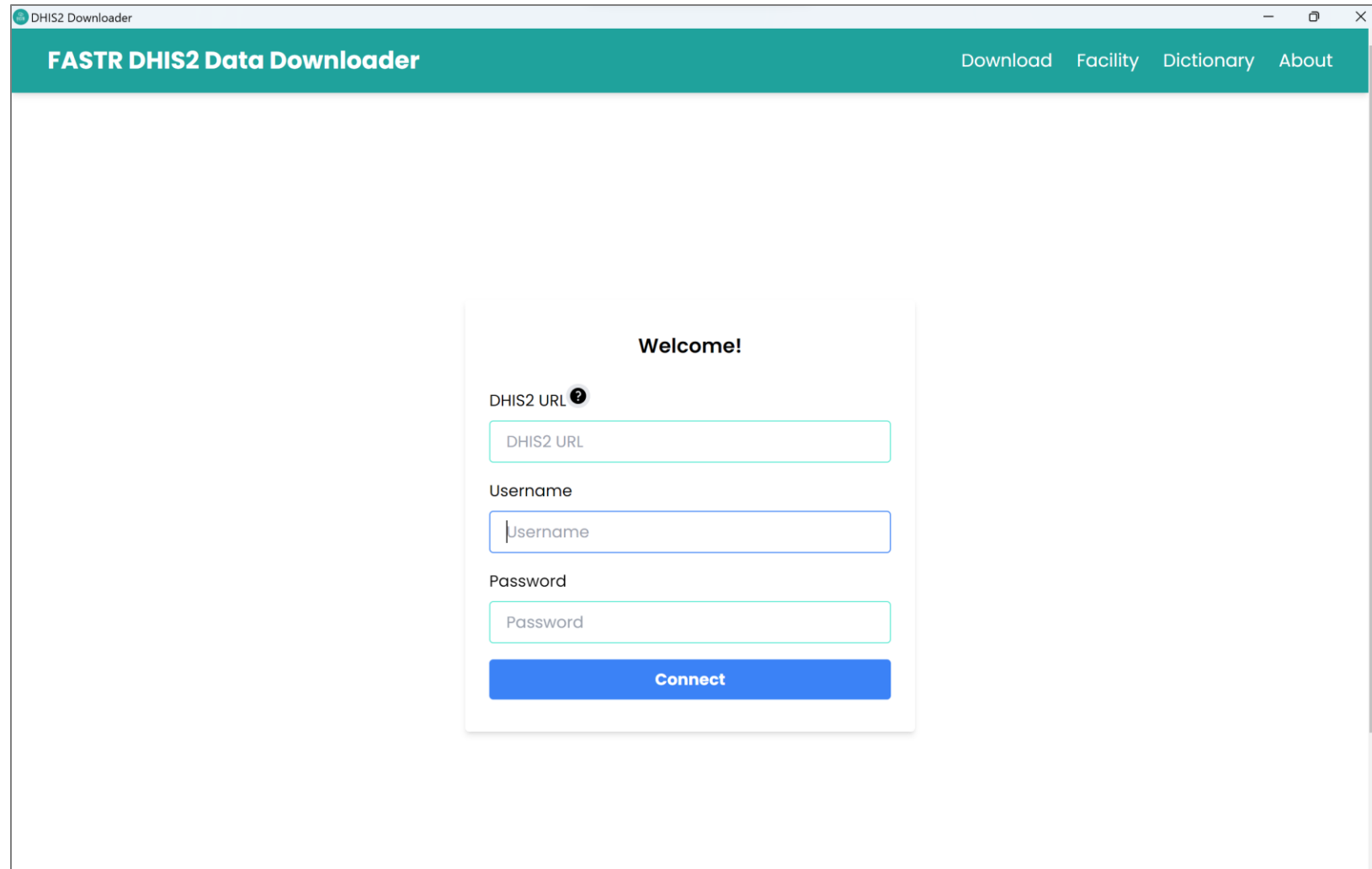
Key features:

- Connect to any DHIS2 instance
- Browse and select data elements and indicators
- Download facility-level data in CSV format
- Maintain download history

Download from GitHub:

<https://github.com/worldbank/DHIS2-Downloader/releases/>

Activity: Data downloader – LIVE Demo and practice



The screenshot shows a web browser window titled "DHIS2 Downloader". The page has a teal header bar with the text "FASTR DHIS2 Data Downloader" on the left and navigation links "Download", "Facility", "Dictionary", and "About" on the right. The main content area is white and features a central login form. The form is titled "Welcome!" and contains three input fields: "DHIS2 URL" (with a help icon), "Username", and "Password". Below these fields is a blue "Connect" button.

DHIS2 Downloader

FASTR DHIS2 Data Downloader Download Facility Dictionary About

Welcome!

DHIS2 URL ?

DHIS2 URL

Username

Username

Password

Password

Connect



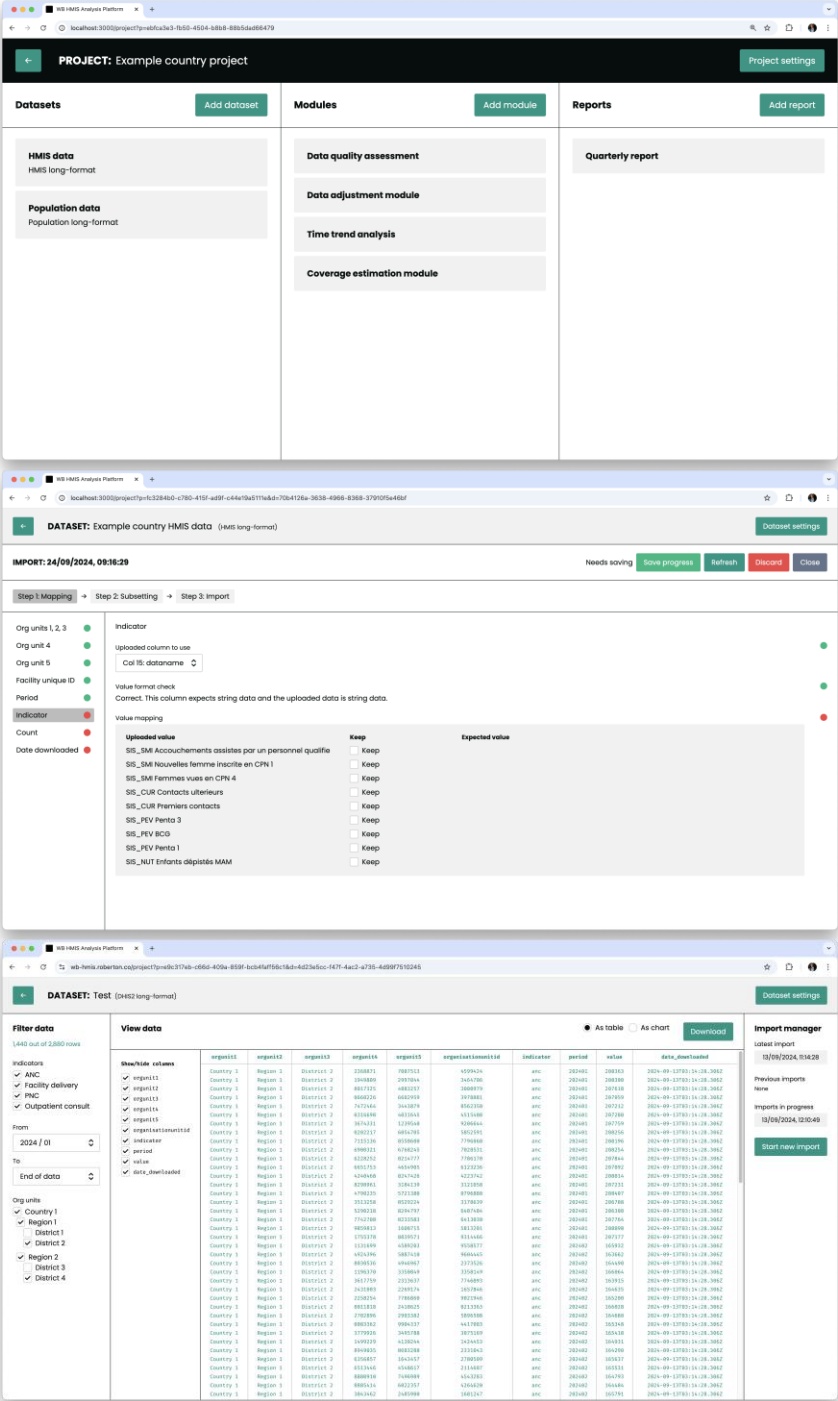
Session 3: Introduction to the FASTR analytics platform

FASTR analytics platform

The FASTR analytics platform is a web-based tool designed to support data quality assessment, adjustment, and analysis for routine health data.

It allows users to upload and analyze data from various sources, including DHIS2, with built-in statistical methods to generate an adjusted dataset and run priority analyses on selected indicators.

The platform provides a user-friendly interface for running analyses and offers flexible options for visualizing and exporting results.



Country Instance

Each country has its own **instance** of the FASTR analytics platform.

An instance contains:

- All registered users and their accounts
- The shared administrative structure (regions, districts, facilities)
- Indicator definitions and data sources
- All projects created for that country

Think of an instance as your country's dedicated workspace.

Projects Within an Instance

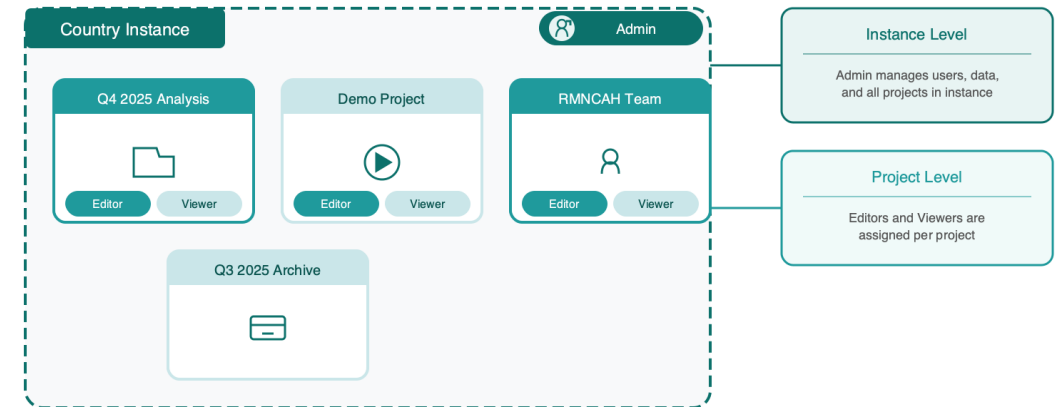
Each country instance can contain **multiple projects**.

A country may only need one project, or multiple projects can be used for:

- Different versions of analyses
- A demo or playground project
- Separate projects for different teams or programs

Key questions when setting up:

- Who is the admin?
- Who can edit?
- Who can view?



User Roles and Permissions

There are two levels of permissions in the platform:

Instance-level roles:

- **Instance Administrators** can add users, create projects, assign roles, upload data, import and configure modules, and run analyses

Project-level roles:

- **Project Editors** can create visualizations, create reports, and download/export results
- **Project Viewers** can view visualizations, view reports, and download/export results

Administrators are assigned per instance; Editors and Viewers are assigned per project.

Practice: Logging Into the Platform



1. Go to <https://zambia.fastr-analytics.org>
2. Click Sign up and enter your details
3. Enter your information (verify email)
4. After login, you'll be added to a project

Create your account

Welcome! Please fill in the details to get started.

First name

Last name

Email address

Password

Continue ▶

Already have an account? [Sign in](#)

Lunch Break

90 minutes • Back at 14:00



Session 4: Configuring the FASTR analytics platform

Configuring the analysis platform

- Configuration of the analysis platform is an admin feature
- We will work together to configure the following items:
 - Admin areas (regions, districts)
 - Facility structure
 - Indicator definitions
- Note since this is an admin feature all participants will NOT be doing this step. Instead, you will select one person to have admin rights, and they will help us walk through these steps.

Activity: Setting Up Admin Areas

- Admin areas (regions, districts)
- Facility structure
- Indicator definitions

Participants will work directly in the platform

Activity: Importing Data

- Review data format requirements
- Walk through the import process
- Handle validation and error checking

Participants will import their country's data

Activity: Installing and Running Modules

- Review available analysis modules
- Install required modules
- Run initial analyses

Participants will configure and run modules on their data

Key messages and wrap-up

REFLECTIONS FROM PARTICIPANTS



STRENGTHENING HEALTH SYSTEMS AND RMNCAH-N OUTCOMES THROUGH RAPID CYCLE ANALYTICS AND DATA USE

*Country Workshop: Introduction to FASTR
RMNCAH-N service Use Monitoring*

Day 2



RECAP OF DAY 1

Agenda

Time	Agenda	Facilitator/Presenter
09:00–09:15	Overview of Day 2 agenda	GFF FASTR team
Session 5: Overview of FASTR methods and analytical outputs		
09:15–10:15	Data quality, service utilization, coverage	GFF FASTR team
Session 6: Creating a project		
10:15–11:15	Project creation and settings	GFF FASTR team
Session 7: Creating visualizations		
11:15–12:30	Creating and editing visualizations	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 8: Creating reports		
14:30–16:30	Practice creating and editing reports	GFF FASTR team

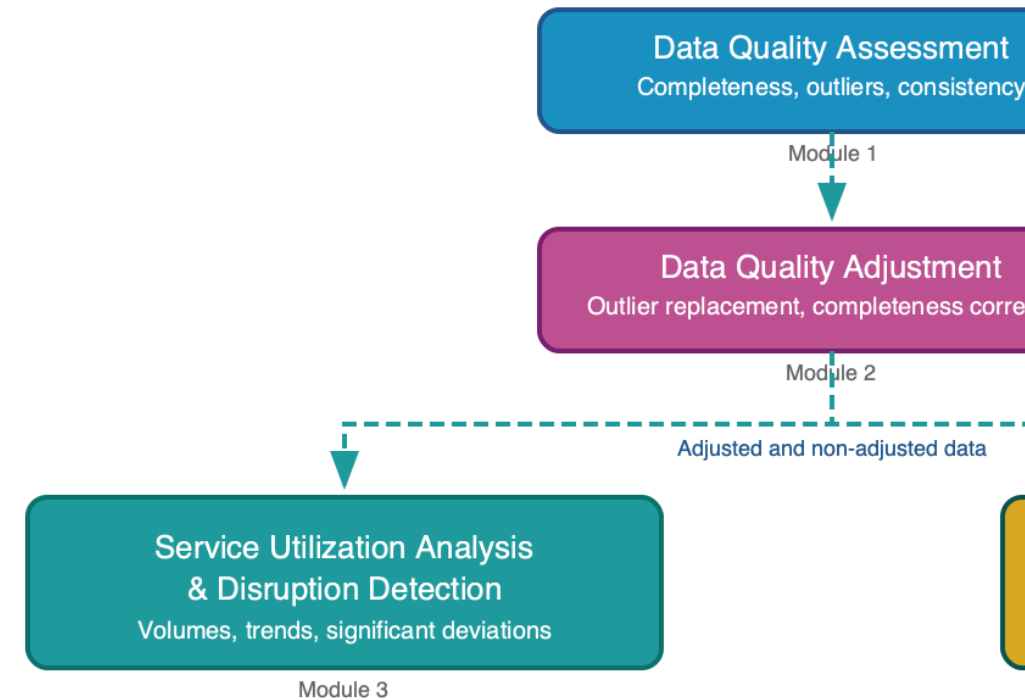


Session 5: Overview of FASTR methods and analytical outputs

FASTR analytical pipeline

The FASTR analysis follows a sequential workflow where each step builds on the previous:

- **Assess data quality** – Identify issues with completeness, outliers, and consistency
- **Adjust for quality issues** – Apply corrections to improve data reliability
- **Analyze adjusted data** – Generate service utilization and coverage estimates



FASTR takes a multi-pronged approach to data quality, with the belief that data quality should not be a barrier to data use – with the right feedback loops, use of data can contribute to improved quality

1. We do granular data quality assessments and adjustments based on facility-level data leveraging HMIS access with an API.
2. We use only high-volume indicators, because the most-used services provide more stable estimates.
3. We focus on variations across time and space rather than specific point estimates and discuss the interpretation and relevance for decision-making with in-country decision-makers.
4. We believe that using the data and providing feedback is the first step to improving the data.

We will discuss each of these areas over the next sessions.

Rationale for data quality assessment

Challenge: Routine health facility data may contain quality limitations:

- Reported values may fall outside plausible ranges
- Reporting gaps affect data completeness
- Inconsistencies exist between related indicators

Implications: Data quality limitations affect decision-making

- Inaccurate assessments of service delivery trends
- Misidentification of areas requiring intervention
- Suboptimal resource allocation

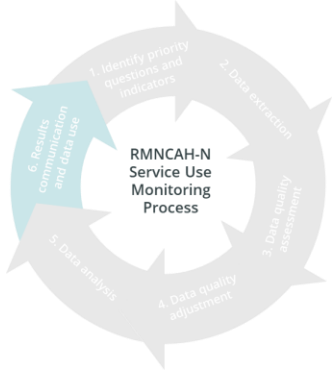
Assessing and adjusting for data quality

Quality assessments identify the highest priority issues and necessary analytical adjustments, so quality issues do not become a barrier to data analysis and use.

Data Quality Summary for Ethiopia Jan 2017 to Oct 2022			
	Completeness	Outliers	Internal Consistency
OPD	Good 80%	Poor 39%	
ANC1	Poor 57%	Good 2%	Good 99%
ANC4	Poor 58%	Good 2%	Good 99%
BCG	Average 73%	Good 1%	Good 93%
Delivery	Good 89%	Good 1%	Poor 6%
Penta 1	Good 84%	Good 1%	Good 95%
Penta 3	Good 83%	Good 1%	Good 95%

Summary ratings are relative across countries participating in monitoring, defines as: **Completeness**: percentage of facilities reporting over the past 12 months; **Outliers**: percent change in volume due to outliers; **Internal consistency**: percent of districts within expected range based on related indicators. Benchmarks are based on scores across multiple countries.

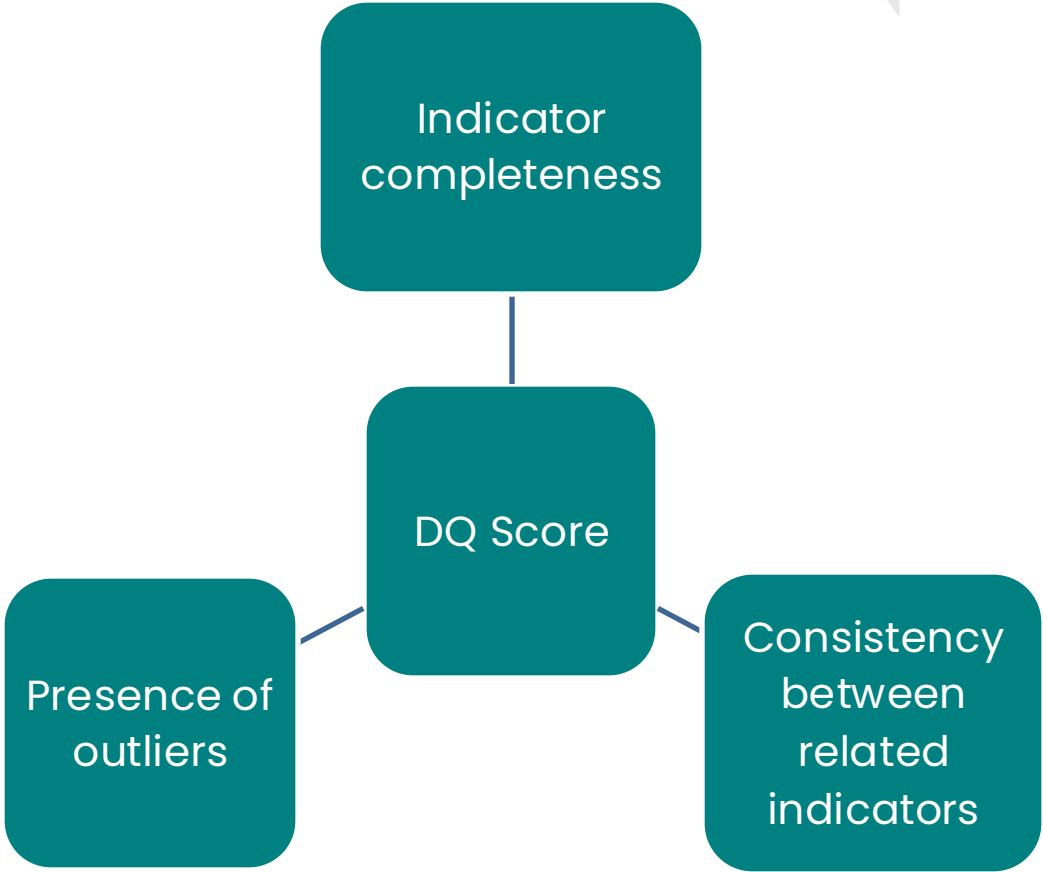
Measures of data quality



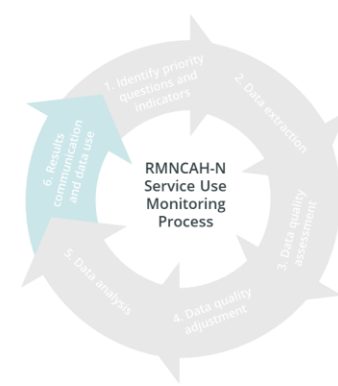
General measures



FASTR measures



How does the FASTR data quality analysis differ from the DQA analysis done in DHIS2?



Purpose of data quality assessment

- DHIS2: focuses on data quality assessment to routinely strengthen data quality over time
- FASTR: focuses on assessing data quality to inform an analysis which answers a pressing policy question

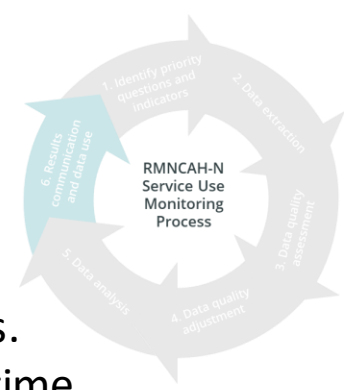
Data quality adjustment

- DHIS2: focus is on identifying data quality issues and working with facilities to improve reporting practices
- FASTR: focus on applying analytical adjustment techniques to account for data quality issues in the analysis; goal is to generate the most robust estimates despite data quality challenges

Selection of indicators, measures, and thresholds – FASTR focuses on DQA elements most relevant for analysis

- The purpose of the data quality assessment guides the selection of indicators, measures, and thresholds.
 - DHIS2 allows configuration of a DQA dashboard for any selection of indicators in DHIS2; FASTR selects indicators that will be used in a specific analysis
 - DHIS2 DQA includes timeliness as a measure of data quality. FASTR does not include timeliness. Timeliness is an important consideration for strengthening routine reporting but it less important to do an analysis with the data we have available today
 - DHIS2 DQA includes reporting completeness (e.g. was a report submitted) and indicator completeness (e.g. was a value recorded for an individual data element) while FASTR focuses only on indicator completeness.

How does the FASTR data quality analysis differ from the DQA analysis done in DHIS2?



Selection of indicators, measures, and thresholds continued

- The purpose of the data quality assessment guides the selection of indicators, measures, and thresholds.
 - DHIS2 DQA assesses four measures of internal consistency: presence of outliers, consistency over time, consistency between related indicators, and consistency between reported data and original records (this metric requires a site assessment / data collection). FASTR focuses on two of these measures: presence of outliers and consistency between related indicators because these are important for analysis and can be done routinely and remotely without visits to health facilities.
 - FASTR and DHIS2 DQA use different outlier detection methods (MADs vs standard deviations); FASTR focuses on identifying VERY large outliers that have undue influence in the analysis and for which adjustments will be made; DHIS2 DQA focuses on identifying outliers that should be followed up at the facility level, with no significant negative impact even if a few correct values are flagged as potential outliers, since these will be investigated further.
 - DHIS2 DQA may assess agreement with external data sources such as periodic population-based surveys and consistency of population data which serves as the denominator for coverage analysis. FASTR does not include this in the data quality assessment but instead incorporates this in our coverage analysis.

Indicator completeness

What it measures: The extent to which facilities report data on selected core indicators

Why it matters:

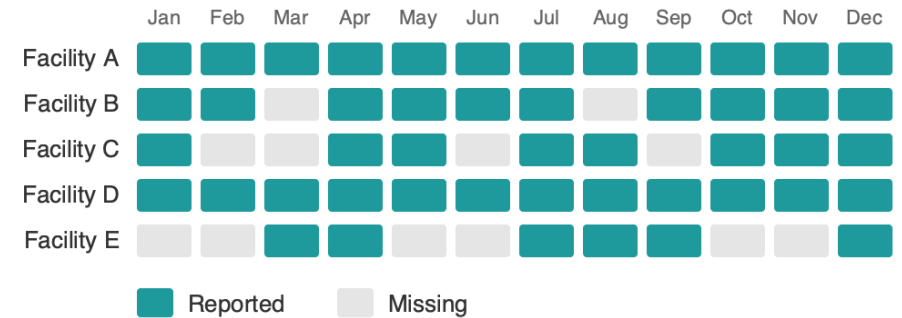
- Higher completeness improves data reliability
- Stability over time strengthens trend analysis

Key distinction:

Indicator completeness $\neq$ reporting completeness. This metric examines specific data elements, not just whether the monthly form was submitted.

Region A: Indicator Completeness

5 health facilities reporting monthly on ANC1



5 facilities $\times$ 12 months = 60 expected
48 reported $\square$ 80% completeness

Definition of indicator completeness

For the FASTR analysis, completeness is defined as:

The percentage of reporting facilities each month out of the total number of facilities expected to report.

- A facility is deemed to be "reporting" if there is a non-missing, non-zero value recorded for the indicator and month
- A facility is expected to report if it has reported any volume for that indicator anytime within a year
- Facilities that do not report for six or more consecutive months at the beginning or end of their reporting period are classified as **inactive** rather than incomplete. This prevents penalizing facilities that have not yet begun reporting or have permanently ceased operations

Notes on completeness

- A high level of completeness does not necessarily indicate that the HMIS is representative of all service delivery in the country as some services may not be delivered in facilities, or some facilities may not report
- For countries where the DHIS2 system does not store 0's, indicator completeness may be underestimated if there are many low-volume facilities for a given indicator

Completeness: Percent of monthly values that are complete

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

For a given indicator in a given time period, the percent of monthly values that are complete:

% complete = # monthly values that are complete / total N of monthly values

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

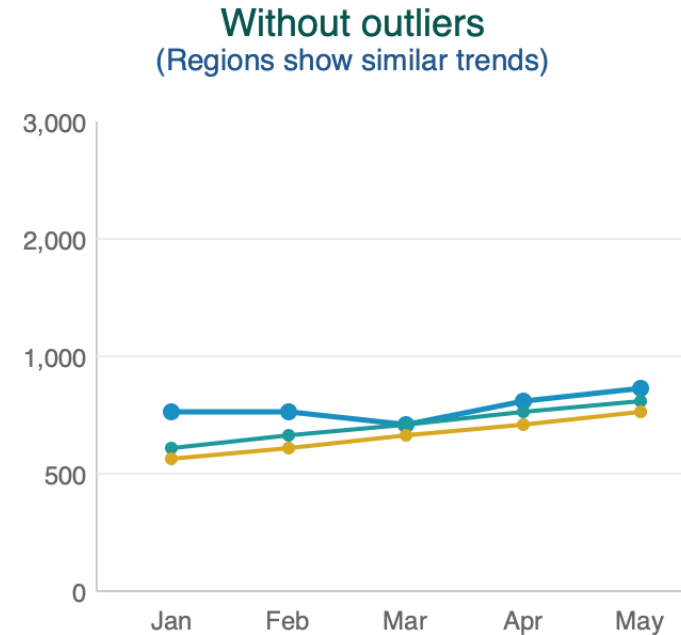
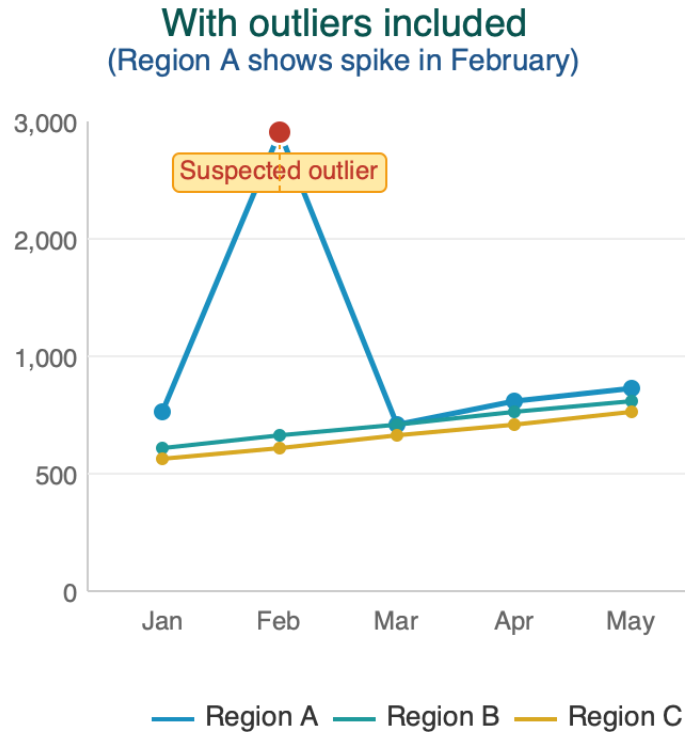
Outliers

The presence of outliers examines whether a data point in a series of values is extreme (either abnormally high or low) in relation to others in the series.

Outliers can be the result of changes in programmatic activities (such as an intensified campaign) or can be data quality problems.

For the FASTR analysis, we identify outliers which are suspiciously high values compared to the usual volume of services reported by the facility (e.g., low values are not identified as outliers in the FASTR analysis).

Outlier illustration



Outliers can distort the true picture of service trends

Region A displays an anomalous spike in February that substantially exceeds values reported by other regions — indicative of a data entry error or reporting issue.

Outlier detection methodology

Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator.

An outlier is defined as:

A value greater than **10 times the median absolute deviation (MAD)** from the monthly median value for the indicator in each time period, **OR** a value for which the proportional contribution in volume for a facility, indicator, and time period is **greater than 80%**

AND for which:

- The volume is **greater than or equal to the median**
- The volume is **not missing**
- The volume is **greater than 100**

Outliers: Percent of monthly values that are outliers

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%

	3% or above
	1% to 2%
	Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

For a given indicator in a given time period, the percent of monthly values that are outliers:

% outliers = # monthly values that are outliers / total N of monthly values

Consistency between related indicators

Program indicators with a predictable relationship are examined to determine whether the expected relationship exists between them. In other words, this process examines whether the observed relationship between the indicators, as shown in the reported data, is that which is expected.

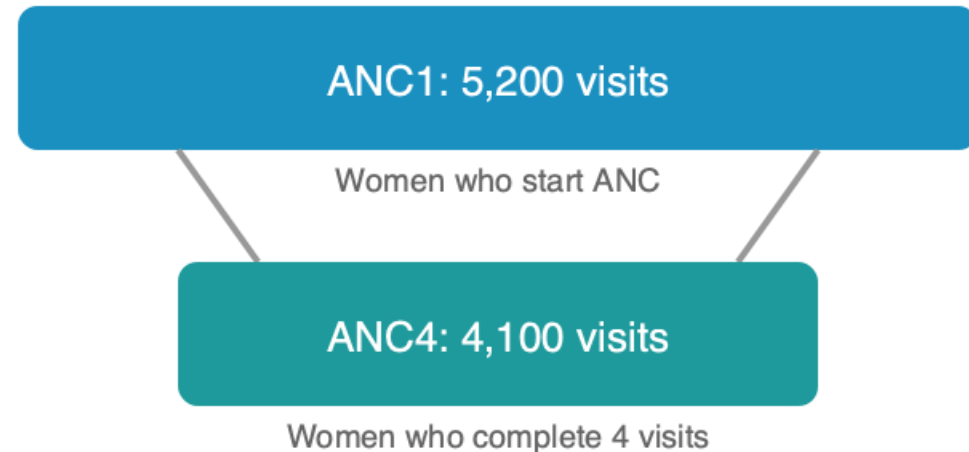
Indicator pairs assessed

Indicator pair	Expected relationship
ANC1 / ANC4	Ratio should be ≥ 0.95
Penta1 / Penta3	Ratio should be ≥ 0.95
BCG / Facility delivery	Within 30% (≥ 0.7 and ≤ 1.3)

These pairs have expected relationships. We expect ANC1 > ANC4 since not all women complete four visits.

BCG is a birth dose vaccine so we expect similar numbers to facility deliveries, with a 30% tolerance for variability.

District X - ANC Services



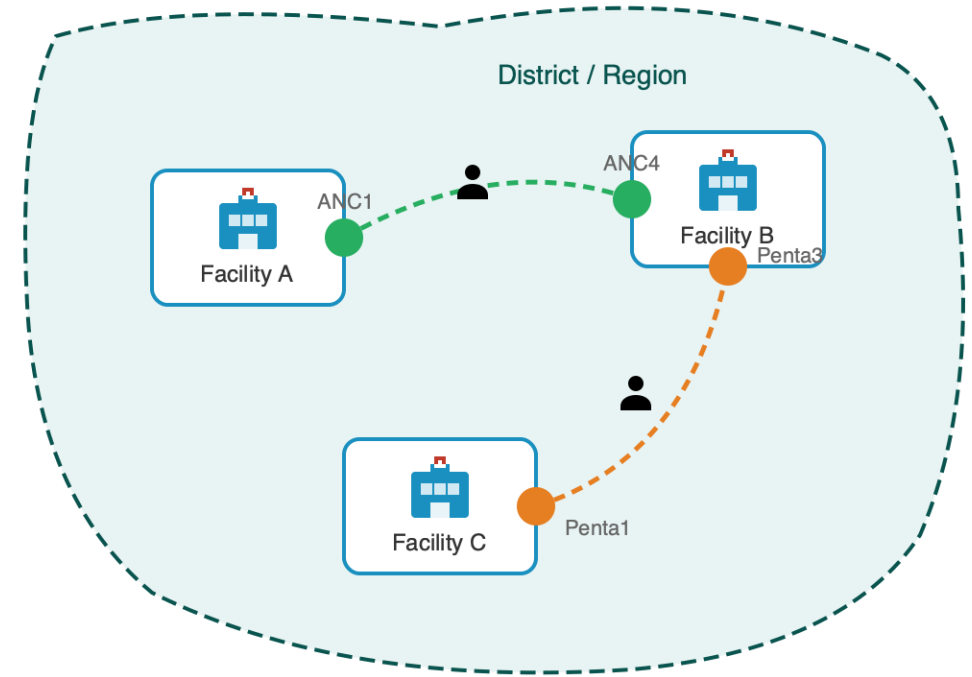
Passes consistency check (ANC1 > ANC4)

Why assess consistency at district level?

Patients often access different services at different facilities within a district:

- A woman may attend **ANC1** at a nearby health post, but travel to a health centre for **ANC4**
- A child may receive **Penta1** at a local clinic, but complete **Penta3** at a district hospital

Checking consistency at the facility level would miss these patterns. Aggregating to district level captures the complete picture of service utilization within a geographic area.



Internal consistency: FASTR output

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

- 90% or above
- 80% to 89%
- Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughly the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

Data quality summary score

A composite measure of data quality provides an overall view of how well a dataset meets quality standards.

By integrating multiple dimensions of data quality into a single score, it simplifies the interpretation of detailed information from several measures. This allows health systems to quickly assess the reliability of data, making it easier to identify trends and issues at a glance.

Definition of adequate data quality

For the FASTR analysis, we defined adequate data quality as:

- No missing indicator data for OPD, Penta1, and ANC1, where available, **AND**
- No outliers for OPD, Penta1, and ANC1, where available, **AND**
- Consistent reporting between Penta1/Penta3 and ANC1/ANC4

Overall DQA score: Percent of monthly values meeting all criteria

Overall DQA score

Percentage of facility-months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

 80% or above
 70% to 79%
 Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

For a given indicator in a given time period, the percent of monthly values meeting all DQA criteria:

% adequate quality = # monthly values meeting all criteria / total N of monthly values

Mean DQA score: How close are we to adequate quality?

Mean DQA score

Average data quality score across facility - months

	2022	2023	2024	2025
Region 001	90.2%	95.5%	95.5%	96.5%
Region 002	87.3%	89.0%	88.2%	85.6%
Region 003	92.8%	94.8%	91.0%	84.3%
Region 004	88.0%	93.9%	93.0%	97.0%
Region 005	89.5%	93.6%	91.1%	88.6%
Region 006	93.2%	96.3%	93.4%	93.4%



Items included in the DQA score include: No missing data for 1) OPD, 2) Penta1, and 3) ANC1, where available; No outliers for 4) OPD, 5) Penta1, and 6) ANC1, where available; Consistent reporting between 7) Penta1/Penta3, 8) ANC1/ANC4, 9)BCG/Delivery, where available.

The mean DQA score shows how close a facility's data is to meeting all quality criteria. A score of **100% means the data passes** all DQA checks—no missing values, no outliers, and consistent reporting.

Mean DQA = (completeness & outlier score + consistency score) / 2

DQA module: Configuration parameters

Parameter	Description
Proportion threshold for outlier detection	Adjusts the threshold for proportional contribution to flag a facility-month as an outlier
Minimum count threshold for consideration	Defines the minimum count required for a facility-month to be considered an outlier
Number of MADs	Outliers are defined as observations which are greater than X times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period
Indicators subjected to DQA	Defines which indicators are included for assessment of outliers and completeness for inclusion in the DQA score
Consistency pairs used	Defines which indicator pairs are used for consistency analysis and the expected ratio ranges

Rationale for data quality adjustment

Routine HMIS data contain two common limitations that can distort analytical results:

Issue	Impact on analysis
Outliers	Extreme values create artificial spikes in service volumes
Incomplete reporting	Missing data creates artificial declines that do not reflect actual service delivery

FASTR addresses these limitations by replacing problematic values with estimates derived from each facility's historical reporting patterns.

Adjustment scenarios

To support transparency and sensitivity analysis, FASTR produces four parallel datasets:

Scenario	Description
Unadjusted	Original reported values
Outliers adjusted	Extreme values replaced
Completeness adjusted	Missing values imputed
Both adjusted	All corrections applied

Indicators excluded from adjustment

Certain indicators are excluded from the adjustment process:

- **Mortality indicators** (maternal deaths, neonatal deaths, under-5 deaths): These represent discrete events where smoothing or imputation is not appropriate
- **Low-volume indicators**: Indicators that never exceed 100 reported events in any month are excluded from adjustment

Outlier adjustment methodology

Outlier values are replaced using facility-specific historical data. The adjustment follows a hierarchical approach:

Priority	Method	Application
1	Centered 6-month average	3 months before + 3 months after the outlier
2	Forward 6-month average	When insufficient preceding data (e.g., start of series)
3	Backward 6-month average	When insufficient following data (e.g., end of series)
4	Same month, previous year	When rolling averages unavailable; useful for seasonal indicators
5	Facility historical mean	Mean of all valid values for this indicator at this facility

Outlier adjustment: FASTR output

Deviance Due to Outliers

Percent change in volume due to outlier adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	0.0%	0.2%	4.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.2%
Bombali District	1.3%	0.0%	0.0%	2.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%
Bonthe District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.5%	0.0%	0.0%	0.0%
Falaba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	2.2%	0.0%	0.0%
Kailahun District	0.0%	0.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	4.5%	2.1%	0.2%	0.2%
Kambia District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Karene District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.7%	1.3%	5.7%	4.7%	0.0%	0.0%
Kenema District	0.0%	0.0%	0.0%	0.0%	0.0%	1.2%	1.7%	1.4%	1.8%	0.4%	0.0%	0.0%
Koinadugu District	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	2.6%	1.9%	0.0%	0.0%
Kono District	0.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.2%	0.6%	2.0%	1.8%
Moyamba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.7%	0.4%	0.1%	0.1%
Port Loko District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Pujehun District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.3%
Tonkolili District	0.0%	0.3%	0.0%	0.0%	0.0%	0.6%	0.0%	0.0%	1.5%	1.4%	0.1%	0.1%
Western Area Rural District	0.0%	0.0%	0.7%	1.5%	1.4%	0.5%	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Western Area Urban District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.6%	0.9%	1.1%	0.5%	1.5%	0.3%

	3% or above
	1% to 2%
	Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100. The deviance is the difference in volume after removing the outlier. High levels of deviance can affect the plausibility of the data.

Completeness adjustment methodology

For months identified as incomplete or missing, values are imputed using the same 6-month rolling average approach applied to outlier adjustment.

Priority	Method	Application
1	Centered 6-month average	When sufficient data exists before and after the gap
2	Forward 6-month average	For gaps at the start of the time series
3	Backward 6-month average	For gaps at the end of the time series
4	Facility historical mean	Mean of all valid values for this indicator at this facility

This approach prevents temporary reporting gaps from creating artificial declines in service volumes.

Completeness adjustment: FASTR output

Deviance Due to Incompleteness

Percent change in volume due to completeness adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	2.2%	4.7%	3.1%	4.7%	3.8%	2.2%	7.2%	9.0%	12.3%	18.4%	3.6%	8.3%
Bombali District	2.6%	3.3%	4.1%	10.4%	12.9%	7.4%	5.9%	5.6%	13.0%	5.6%	10.8%	8.6%
Bonthe District	1.5%	5.2%	3.6%	8.3%	4.3%	1.6%	4.1%	4.0%	24.3%	8.3%	2.7%	6.1%
Falaba District	0.5%	2.8%	3.0%	6.4%	4.2%	0.9%	0.3%	0.3%	12.8%	7.5%	1.8%	4.7%
Kailahun District	0.5%	1.9%	1.3%	13.8%	11.9%	0.5%	0.4%	0.4%	8.3%	8.1%	2.8%	4.4%
Kambia District	0.8%	2.8%	0.4%	1.4%	1.3%	0.5%	0.6%	0.8%	9.6%	5.3%	0.3%	1.4%
Karene District	0.3%	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	8.3%	6.3%	2.3%	4.8%
Kenema District	0.3%	1.2%	0.7%	1.7%	1.3%	0.1%	0.2%	0.2%	5.6%	4.5%	0.5%	1.1%
Koinadugu District	2.3%	3.9%	2.7%	4.3%	2.6%	1.5%	2.1%	2.3%	41.1%	23.9%	13.1%	14.9%
Kono District	0.4%	2.1%	1.2%	5.6%	3.1%	0.6%	0.7%	0.9%	6.8%	7.3%	1.2%	6.5%
Moyamba District	1.4%	3.8%	2.9%	4.0%	3.3%	1.7%	2.0%	2.2%	13.9%	7.4%	2.8%	5.3%
Port Loko District	1.5%	2.0%	2.1%	2.9%	4.8%	1.6%	1.3%	1.4%	19.9%	9.8%	10.3%	8.8%
Pujehun District	0.3%	1.1%	2.1%	5.2%	4.0%	0.1%	0.2%	0.3%	7.0%	3.9%	0.3%	0.9%
Tonkolili District	1.2%	3.0%	4.6%	6.5%	2.7%	1.5%	1.8%	1.7%	9.6%	8.2%	2.5%	3.7%
Western Area Rural District	0.7%	0.8%	1.7%	1.7%	2.0%	0.6%	2.6%	2.4%	22.7%	10.1%	0.2%	0.6%
Western Area Urban District	6.2%	8.6%	5.7%	15.9%	13.5%	5.7%	6.6%	6.4%	19.9%	15.5%	4.5%	6.4%

	3% or above
	1% to 2%
	Below 1%

Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. The deviance is the difference in volume after imputing incomplete data. High levels of deviance can affect the plausibility of the data.

Combined adjustment: FASTR output

Deviance Due to Incompleteness and Outliers

Percent change in volume due to both outlier and completeness adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	2.2%	4.5%	1.5%	4.7%	3.8%	2.2%	7.2%	9.0%	12.3%	18.4%	3.4%	8.1%
Bombali District	1.3%	3.3%	4.1%	8.1%	12.9%	7.4%	5.9%	5.6%	13.0%	4.7%	10.8%	8.6%
Bonthe District	1.5%	5.2%	3.6%	8.3%	4.3%	1.6%	4.1%	4.0%	21.8%	8.3%	2.7%	6.1%
Falaba District	0.5%	2.8%	3.0%	6.4%	4.2%	0.9%	0.4%	0.3%	12.8%	5.3%	1.8%	4.7%
Kailahun District	0.5%	1.9%	0.7%	13.8%	11.9%	0.5%	0.4%	0.4%	3.8%	5.9%	2.6%	4.2%
Kambia District	0.8%	2.8%	0.4%	1.4%	1.3%	0.5%	0.6%	0.8%	8.8%	5.3%	0.3%	1.4%
Karene District	0.3%	0.8%	0.0%	0.0%	0.0%	0.0%	2.7%	1.0%	2.6%	1.6%	2.3%	4.8%
Kenema District	0.3%	1.2%	0.7%	1.7%	1.3%	1.1%	1.5%	1.2%	3.7%	4.1%	0.5%	1.1%
Koinadugu District	2.3%	3.9%	2.7%	4.3%	2.3%	1.5%	2.1%	2.3%	38.6%	22.1%	13.1%	14.9%
Kono District	0.3%	2.1%	1.2%	5.6%	3.1%	0.6%	0.7%	0.9%	3.6%	6.8%	0.9%	4.7%
Moyamba District	1.4%	3.8%	2.9%	4.0%	3.3%	1.7%	2.0%	2.2%	10.2%	7.0%	2.7%	5.2%
Port Loko District	1.5%	2.0%	2.1%	2.9%	4.8%	1.6%	1.3%	1.4%	19.1%	9.8%	10.3%	8.8%
Pujehun District	0.3%	1.1%	2.1%	5.2%	4.0%	0.1%	0.2%	0.3%	7.0%	3.9%	0.0%	0.6%
Tonkolili District	1.2%	2.7%	4.6%	6.5%	2.7%	0.9%	1.8%	1.7%	8.1%	6.7%	2.4%	3.6%
Western Area Rural District	0.7%	0.8%	0.9%	0.2%	0.6%	0.1%	1.6%	2.4%	22.7%	10.1%	0.2%	0.6%
Western Area Urban District	6.2%	8.6%	5.7%	15.9%	13.5%	5.7%	6.0%	5.5%	18.7%	15.0%	3.0%	6.1%

	3% or above
	1% to 2%
	Below 1%

TBD

Service utilization methods

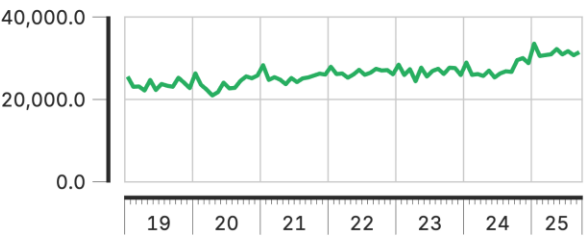
- Assessing trends in service utilization allows us to track changes in the volume of priority health services over time, providing a direct measure of service delivery. The goal is to detect patterns and trends across time to highlight areas where fluctuations may indicate disruptions or reforms affecting utilization.
- We assess:
 - Trends over time
 - Absolute yearly volume for selected services
 - Percent change in volume
 - Any year with more than a 10% change compared to the previous year is flagged, as significant shifts in service volume may indicate disruptions or reforms affecting utilization, or they could signal potential artifacts resulting from changes in data quality.
- Data can be adjusted for outliers only, completeness only, or both outliers and completeness (we will see how to change this in the visualizations).

Service utilization over time

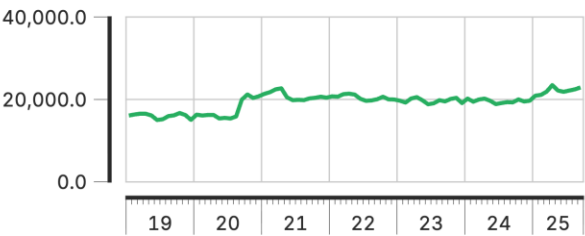
Service utilization over time

Jan 2019 to Sep 2025

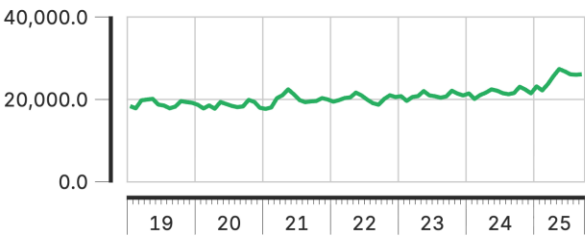
Antenatal client 1st visit



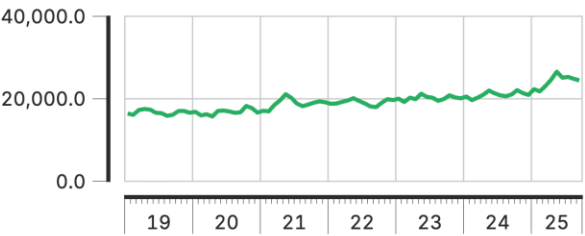
Antenatal client 4th visit



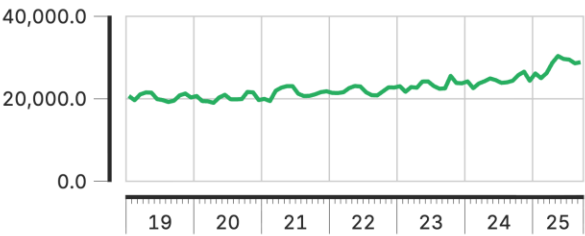
Institutional delivery (normal, assisted, and csection)



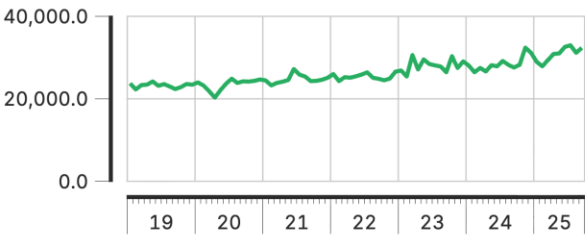
Postnatal care 1 (mothers)



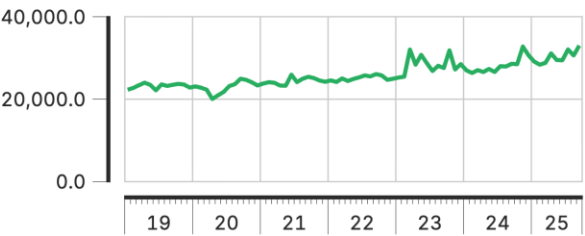
BCG dose



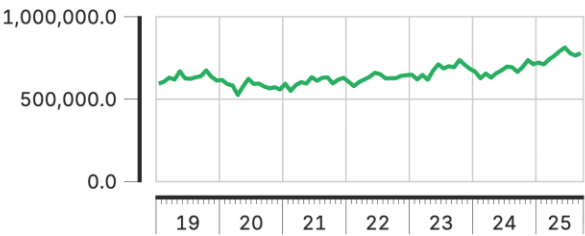
Pentavalent 1st dose



Pentavalent 3rd dose



Outpatient visit



Number of services after both outlier and completeness adjustment

Service volume is adjusted for outliers and completeness.

Year-over-year change

Year-over-year percent change quantifies shifts in service delivery between consecutive years.

For each indicator and region, total volume in the current year is compared to the previous year:

Percent change = $(\text{Current year} - \text{Previous year}) \div \text{Previous year} \times 100$

Changes exceeding **±10%** are flagged for review.

Output: Change in service volume

Interpretation

Bars show annual service volumes by region.
Percentages indicate year-over-year change.

Key considerations:

- Which regions exhibit the largest changes?
- Are changes consistent across regions or geographically concentrated?
- Do patterns vary by indicator?

Service volume by year & year-on-year change

Jan 2019 to Sep 2025



Service utilization comparison to DHIS2

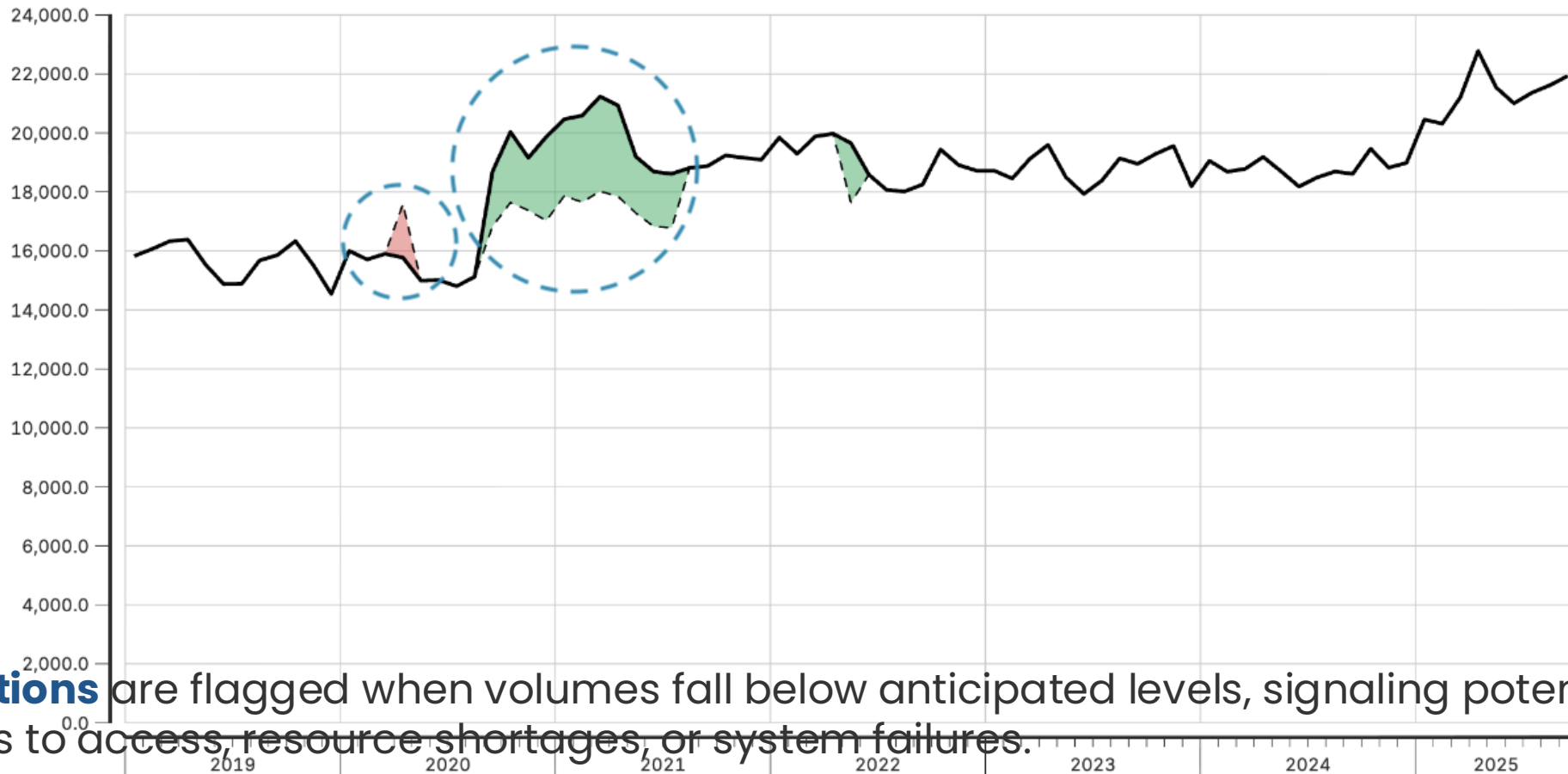
- Service utilization trends are commonly produced in DHIS2
- Main differences between DHIS2 and FASTR approach
 - FASTR approach adjusts for data quality (outliers and/or completeness)
 - FASTR approach visualizes data with the percent change approach to make it easier to identify meaningful fluctuations in service delivery
 - FASTR approach looks at trends, but also uses the adjusted service utilization data for additional, more complex analyses

Disruption detection

Our approach to service disruptions and surpluses utilizes an interrupted time series regression with facility-level fixed effects. Previous large and unexpected changes in historical data are removed. Unexpected volume changes are estimated by comparing observed volume to expected volume based on historical trends and seasonality.

Disruptions and surpluses

Antenatal client 4th visit



Disruptions are flagged when volumes fall below anticipated levels, signaling potential barriers to access, resource shortages, or system failures.

Surpluses occur when volumes exceed expectations, which may indicate increased demand, over-reporting, or changes in service delivery.

How it works

Using past data to set expectations: We look at the past few years of data to understand the typical pattern for each month, accounting for regular seasonal changes.

Spotting unusual changes: We compare current service volumes to expectations. If we see volumes much higher or lower than expected, we flag it as an unusual change.

Handling past disruptions: We adjust historical data by removing previous big, unexpected changes so one-off events don't skew our understanding of what's "normal."

Detecting disruptions over time: We look at trends to see if there are clear shifts in health service use over several months.

Comparison to DHIS2

Extension of service utilization analysis, using more complex statistical approaches not available in DHIS2.

Using a regression framework, we are able to:

- Account for seasonality
- Exclude unusual changes to ensure one-off events aren't influencing normal trends
- Use historical data as a baseline for context
- Detect disruptions and recovery patterns
- Quantify changes with a robust methodology as compared to just observing simple fluctuations in a trend line

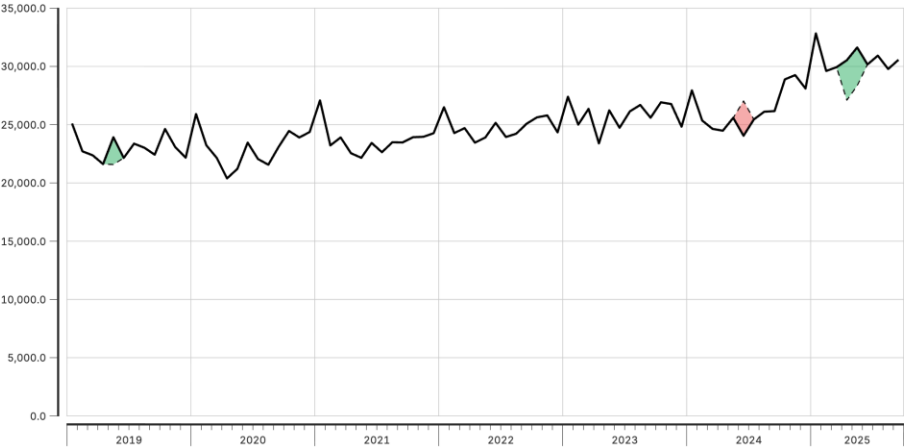
This improves the ability to interpret and compare utilization data across national and sub-national areas without needing population denominators.

Output: Actual vs expected (national)

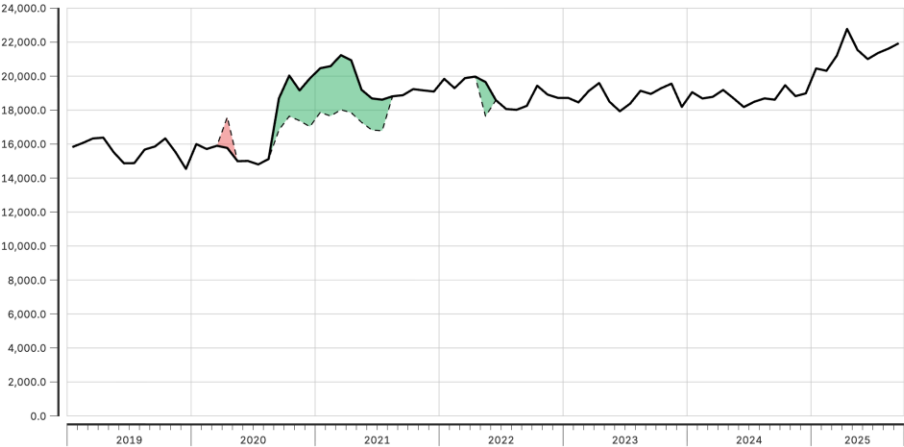
Comparing reported service use to expected trends, nationally

Jan 2019 to Sep 2025

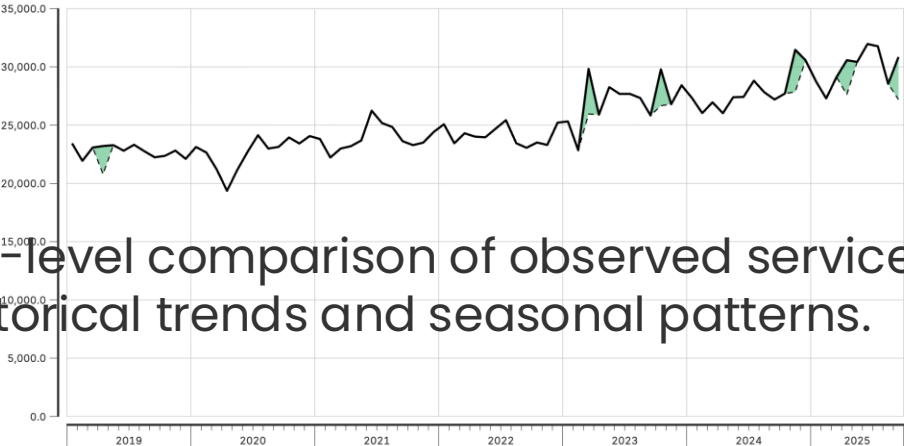
Antenatal client 1st visit



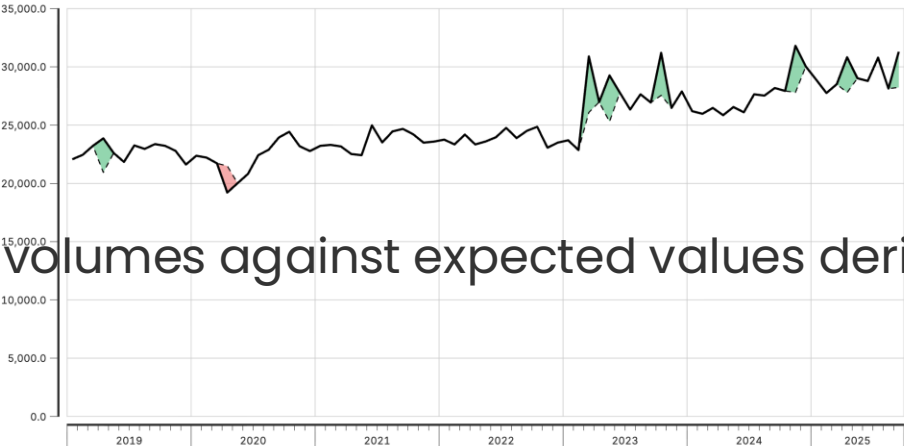
Antenatal client 4th visit



Pentavalent 1st dose



Pentavalent 3rd dose



National-level comparison of observed service volumes against expected values derived from historical trends and seasonal patterns.

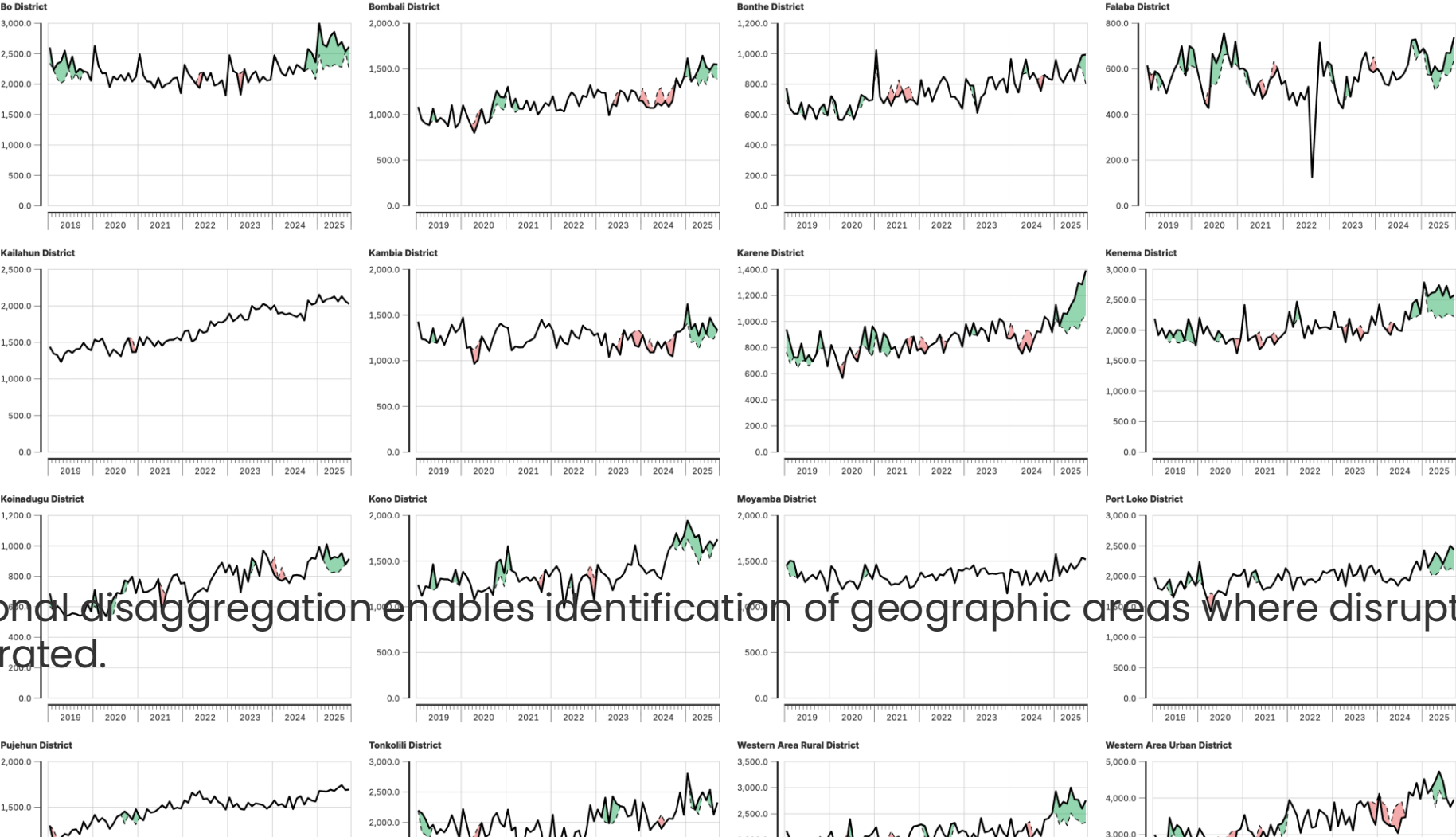
— Actual
- - - Expected
Surplus
Disruption

This graph quantifies changes in service volume compared to historical trends and accounting for seasonality. These signals should be triangulated to other data and contextual knowledge to determine if the results are an artifact of data quality. Unexpected volume changes are estimated by comparing the observed volume to the expected volume based on historical trends and seasonality. Previous large unexpected changes in the historical data are removed. This analysis is an interrupted time series regression with facility-level fixed effects.

Output: Actual vs expected (subnational)

Comparing reported service use to expected trends, sub - nationally

Jan 2019 to Sep 2025



Subnational disaggregation enables identification of geographic areas where disruptions are concentrated.

Service utilization module: Configuration parameters

Parameter	Description
Count variable to use for modeling	Which adjusted count to use for calculating expected values
Count variable to use for visualization	Which adjusted count to use as actual observed values
Run district-level model	Run regressions at admin_area_3 level. Set to Yes for detailed analysis, No for faster runtime
Run admin_area_4 analysis	Run finest-level analysis. Warning: can be very slow for large datasets
Threshold for MAD-based control limits	Number of MADs to flag sharp deviations. Default 1.5; higher = less sensitive
Smoothing window (k)	Window size in months for rolling median smoothing. Must be odd. Default 7
Dip threshold	Flag if actual falls below this proportion of expected. Default 0.9 ($\geq 10\%$ drop); use 0.8 to flag only big drops
Difference percent threshold	Highlight points where actual differs from expected by more than this percent. Default 10

Service coverage estimates – Module 4

Estimating the percentage of the target population that received a given health service

Our approach to service coverage analysis

Our approach derives and validates population denominators, significantly improving coverage estimates reported from HMIS systems.

In countries with accurate data, this approach helps identify subnational inequities and updates outdated estimates, while in countries with less precise data, the trends still provide valuable insights into performance.

We use these estimates to track recent trends and subnational disparities in the coverage of selected health services.

Two-part analytical process

The coverage estimation module operates in two sequential parts:

Part	Components
Part 1: Denominator calculation	Calculate target populations using multiple methods; compare against survey benchmarks; select optimal denominator for each indicator
Part 2: Coverage estimation	Apply denominator selections; project survey estimates forward using HMIS trends; generate final coverage estimates

What is service coverage?

$$\text{Service coverage} = \frac{\text{Population who received the service}}{\text{Population who need the service (target population)}}$$

Numerator comes from DHIS2 data, count of services adjusted for outliers

Several options for estimating the denominator

Service coverage represents the proportion of the target population that received a specified health service.

Denominators by service type

Each health indicator corresponds to a specific target population:

Service	Target population (denominator)
ANC1, ANC4	Pregnant women
Institutional delivery	Live births
BCG	Live births
Penta1, Penta3	Infants surviving beyond neonatal period
Measles1, Measles2	Infants surviving beyond infancy

Service coverage example

Number of pregnant women who
had 4+ ANC (from DHIS-2)



#ANC4+

Coverage of ANC4+ =

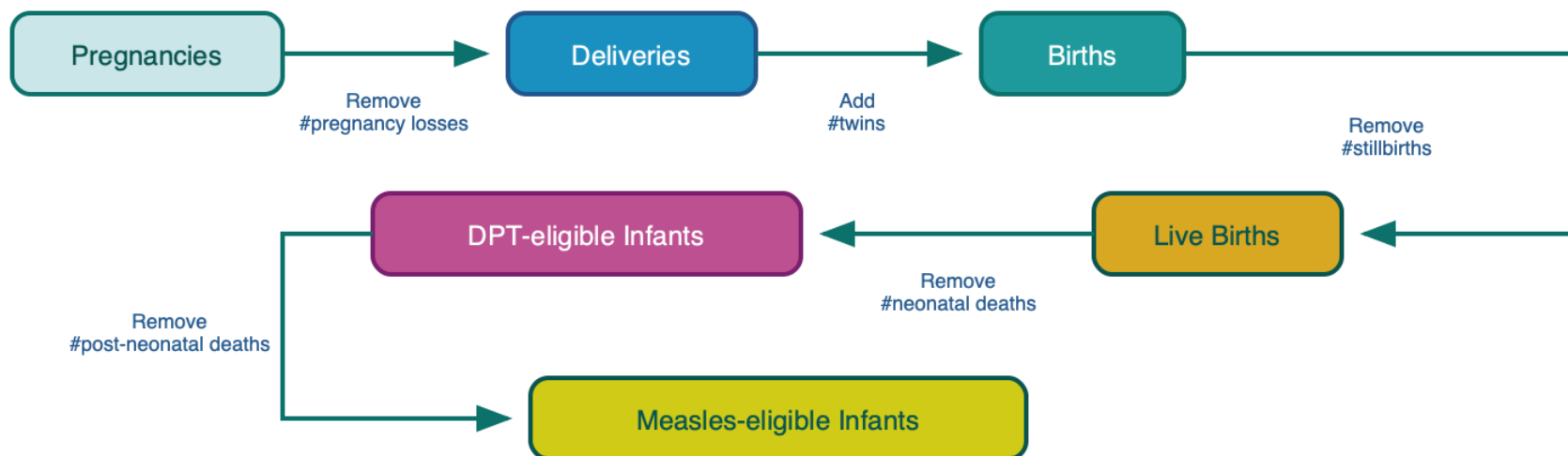


#Pregnancies



Total pregnancies in the population
(not available in DHIS-2)

Demographic cascade



Default values:

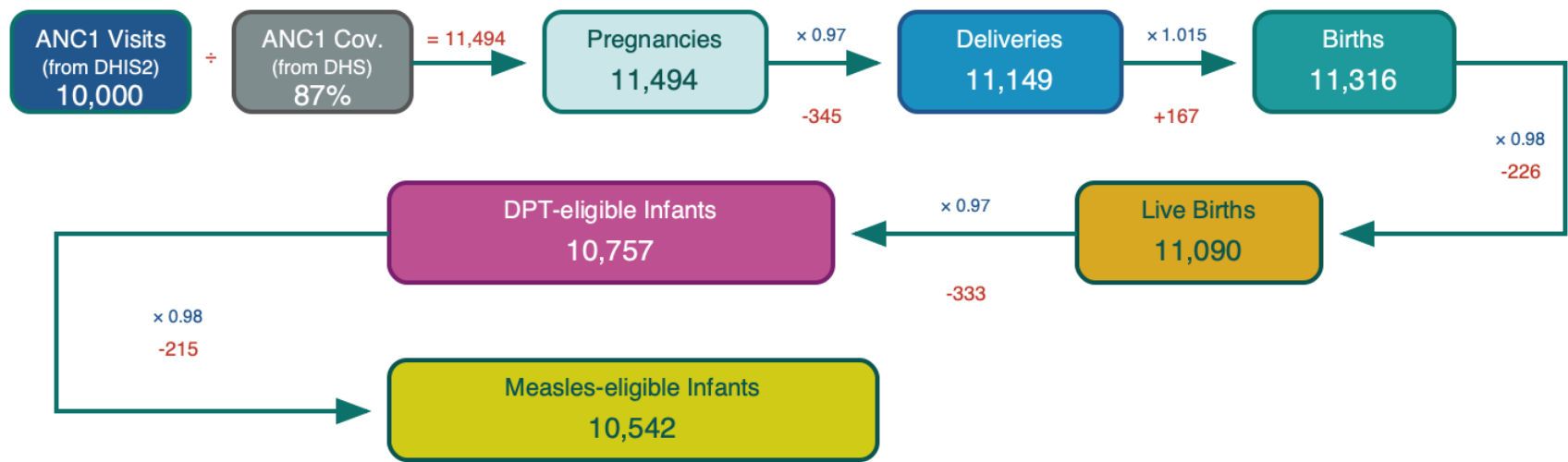
Pregnancy loss rate = 0.03 | Twin rate = 0.015 | Stillbirth rate = 0.02 | NMR = 0.03 | PNMR = 0.02

Use country-specific values for stillbirth rate, NMR, PNMR, and IMR when available from DHS/MICS.

Sequential demographic adjustments transform one target population estimate into another. Starting from pregnancies, demographic factors are applied to derive subsequent denominators:

Denominator cascade: Illustration

Example: Calculating denominators from ANC1 visits



Calculation summary:

Step 1: Pregnancies = ANC1 visits ÷ ANC1 coverage = 10,000 ÷ 0.87 = 11,494

Step 2: Apply demographic factors through the cascade:

Pregnancies (11,494) □ Deliveries (11,149) □ Births (11,316) □ Live Births (11,090) □ DPT-eligible (10,757) □ Measles-eligible (10,542)

Rates: Pregnancy loss = 3%, Twin rate = 1.5%, Stillbirth = 2%, NMR = 3%, PNMR = 2%

Starting from ANC1 service counts, demographic adjustment factors are applied sequentially to derive denominators for other services:

Forward and backward derivation

From any entry point, the cascade derives denominators in both directions:

Direction	Method	Example from Penta1
Forward	Apply mortality/attrition rates	DPT-eligible → Measles1-eligible → Measles2-eligible
Backward	Reverse mortality rates (add deaths back)	DPT-eligible → Live births → Births → Deliveries → Pregnancies

Backward derivation enables estimation of upstream populations from downstream service counts.

Denominators by entry point

Each HMIS indicator serves as an entry point. The module derives all target populations via forward and backward cascades:

Entry point	Base calculation	Forward derivation	Backward derivation
ANC1	ANC1 ÷ coverage → Pregnancies	Deliveries → Live births → DPT-eligible → Measles-eligible	—
Deliveries	Deliveries ÷ coverage → Deliveries	Live births → DPT-eligible → Measles-eligible	Pregnancies
BCG	BCG ÷ coverage → Live births	DPT-eligible → Measles-eligible	Deliveries → Pregnancies
Penta1	Penta1 ÷ coverage → DPT-eligible	Measles1-eligible → Measles2-eligible	Live births → Births → Deliveries → Pregnancies
UN WPP	Crude birth rate × population → Pregnancies, live births; Under-1 pop → DPT, measles	Applies mortality rates for measles denominators	—

Five denominator options for FASTR analysis

1. UN projections of live births (available only at National level)

Service utilization (DHIS2) derived denominators

1. Adjustment to ANC1 service utilization
2. Adjustment to delivery service utilization
3. Adjustment to BCG service utilization (only calculated at national level)
4. Adjustment to Penta1 service utilization (only calculated at national level)

Automatic denominator selection

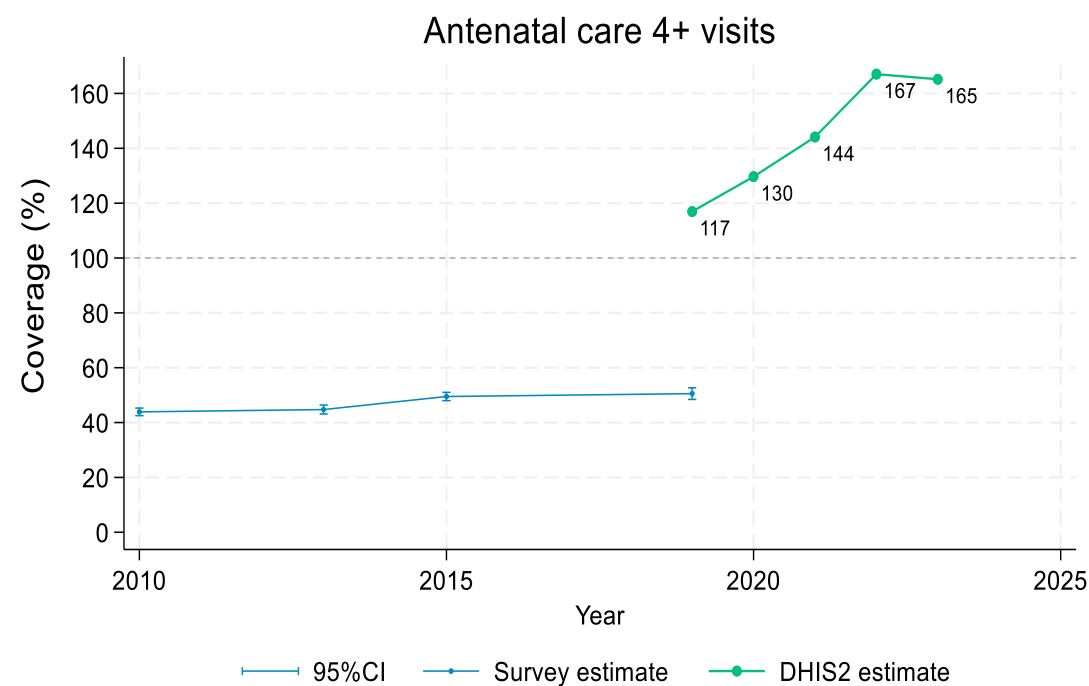
For each indicator, the module selects the denominator that produces coverage closest to the survey benchmark.

Selection algorithm:

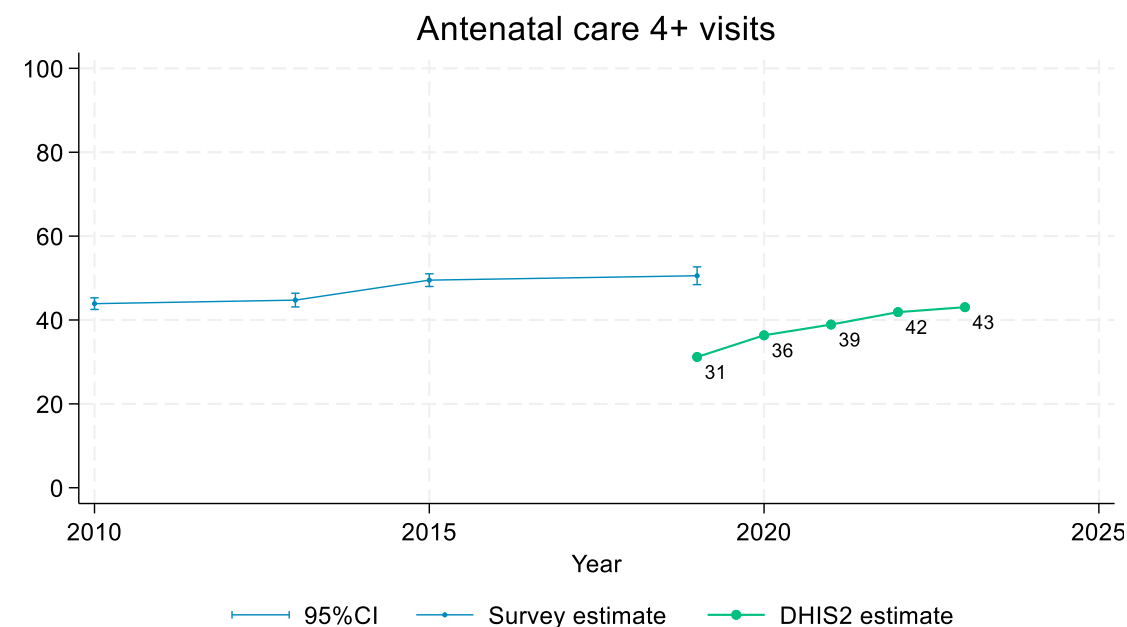
- Calculate coverage using each denominator option
- Calculate squared error against survey: $(\text{coverage} - \text{survey})^2$
- Apply selection hierarchy (HMIS-based denominators prioritized over UN WPP)
- Select the HMIS-based denominator with minimum error

Selection is made per indicator and geographic area. Users may override automatic selections in Part 2.

Which denominator provides estimates that are more plausible and closer to the survey estimates?

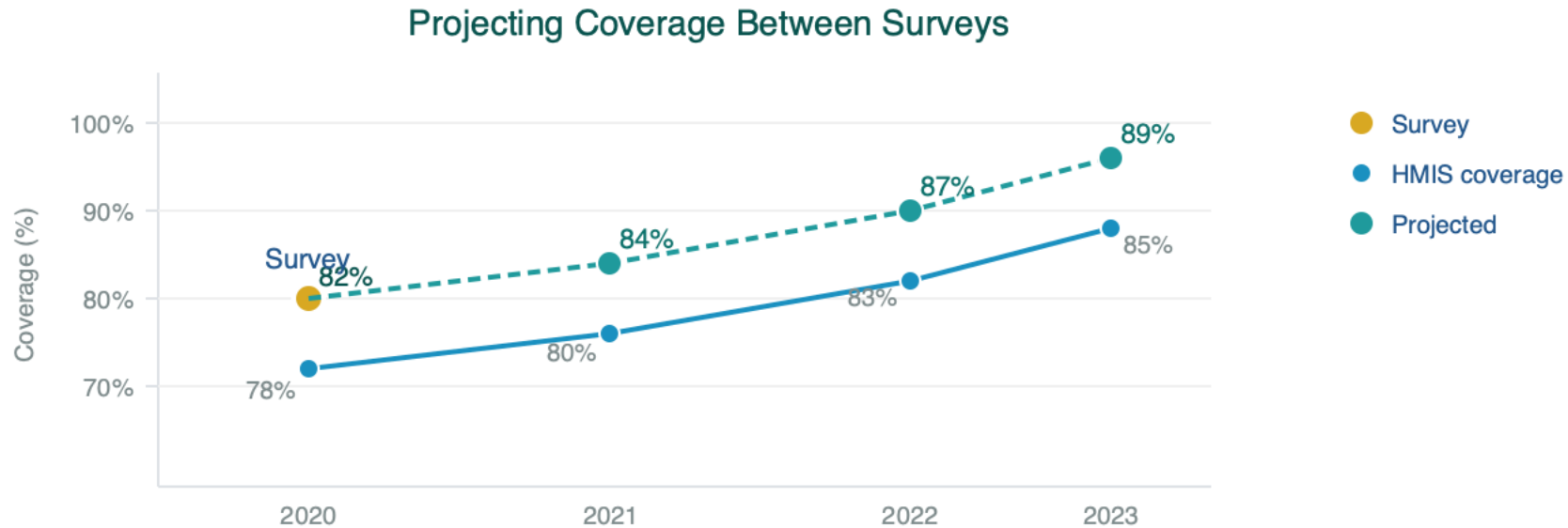


Denominators derived from ANC1 estimates



Denominators derived from UN projected live births

Coverage projection methodology



The module projects the most recent survey value forward using trends observed in HMIS-derived coverage:

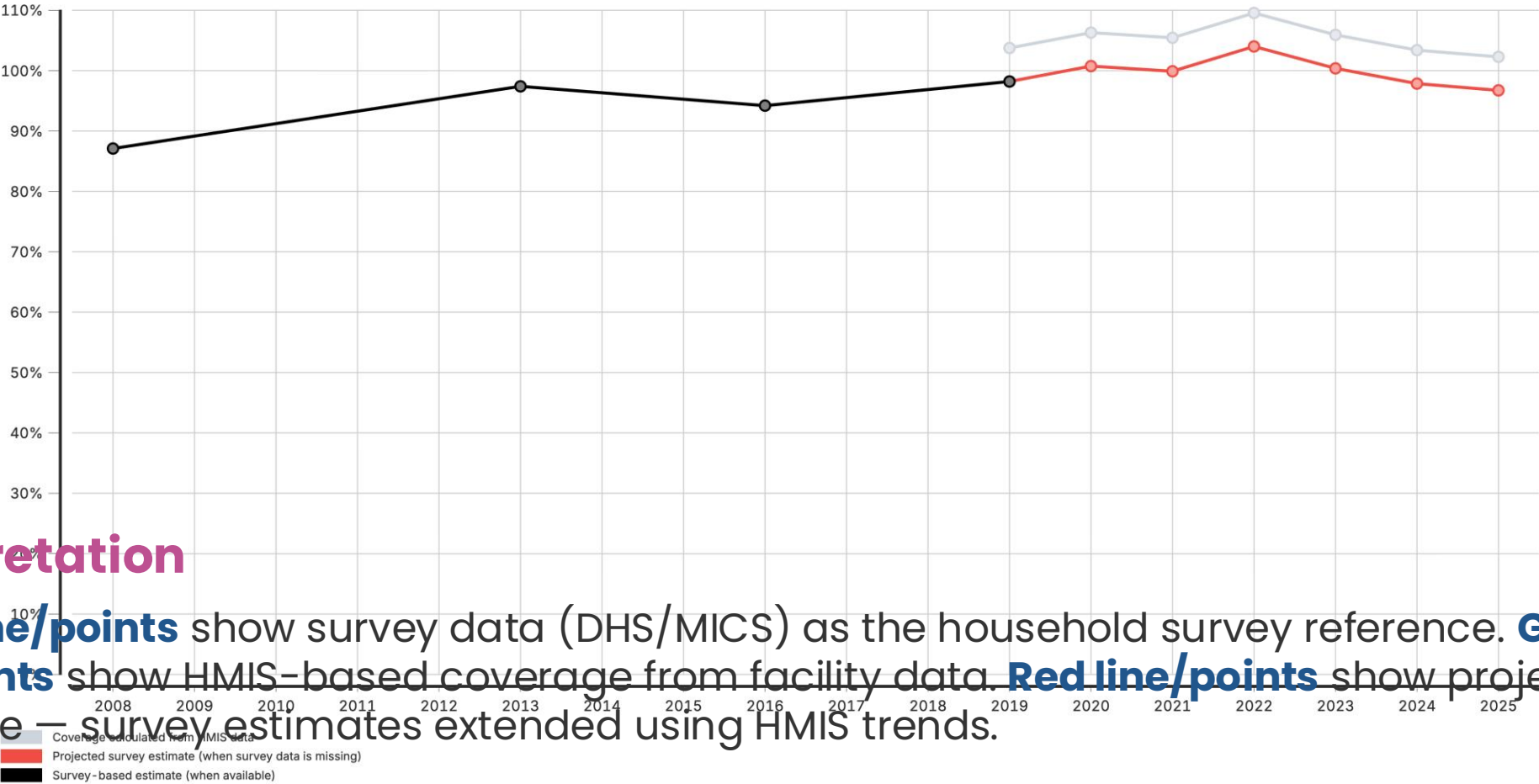
Year-over-year changes (deltas) in HMIS coverage are calculated and applied to the last survey value. This approach preserves the survey baseline while incorporating observed service delivery trends.

Coverage (national)

Coverage estimates for Antenatal client 1st visit

2008 to 2025

DISCLAIMER: These results use routine data to provide rigorous, but not official estimates. They should be interpreted considering any data quality or representation limitations, including data quality findings and any other country specific factors.



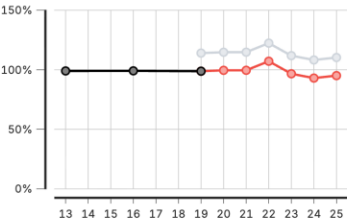
Coverage (subnational)

Subnational coverage estimates for Antenatal client 1st visit

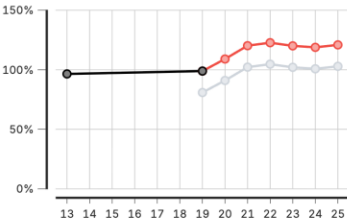
2013 to 2025

DISCLAIMER: These results use routine data to provide rigorous, but not official estimates. They should be interpreted considering any data quality or representation limitations, including data quality findings and any other country specific factors.

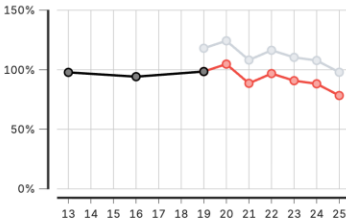
Bo District



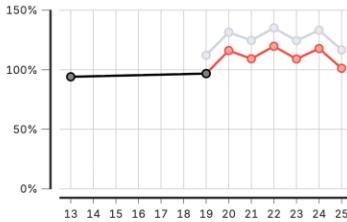
Bombali District



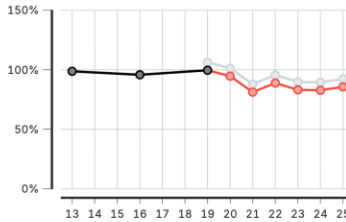
Bonthe District



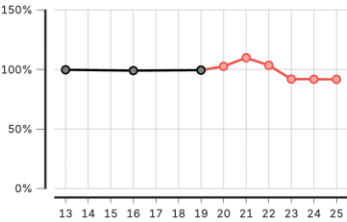
Falaba District



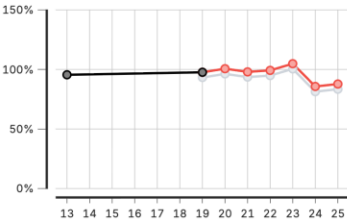
Kailahun District



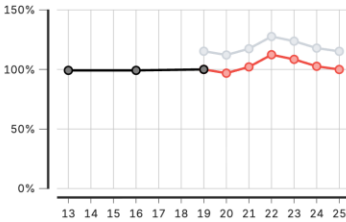
Kambia District



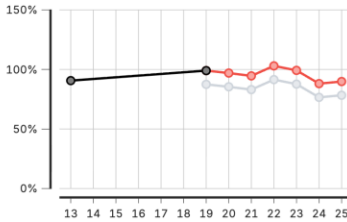
Karene District



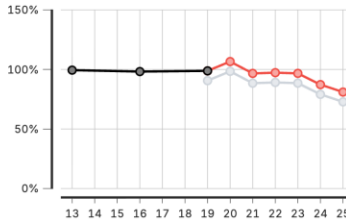
Kenema District



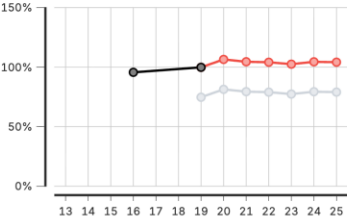
Koinadugu District



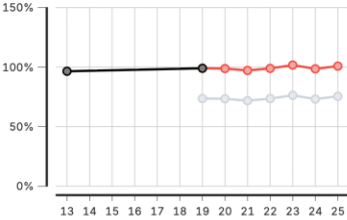
Kono District



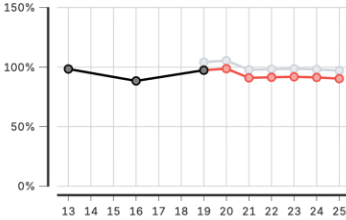
Moyamba District



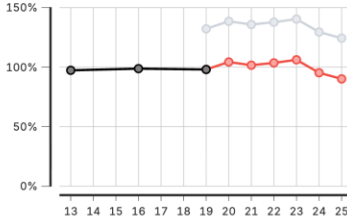
Port Loko District



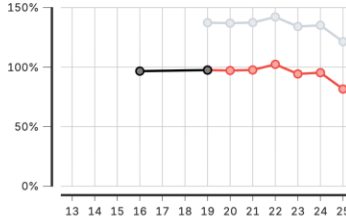
Pujehun District



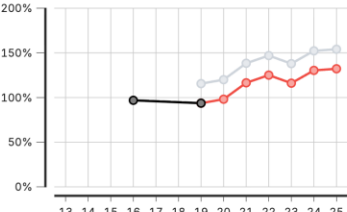
Tonkolili District



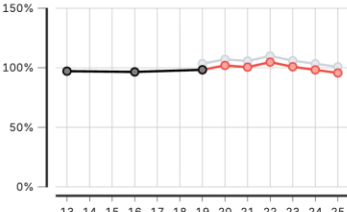
Western Area Rural District



Western Area Urban District



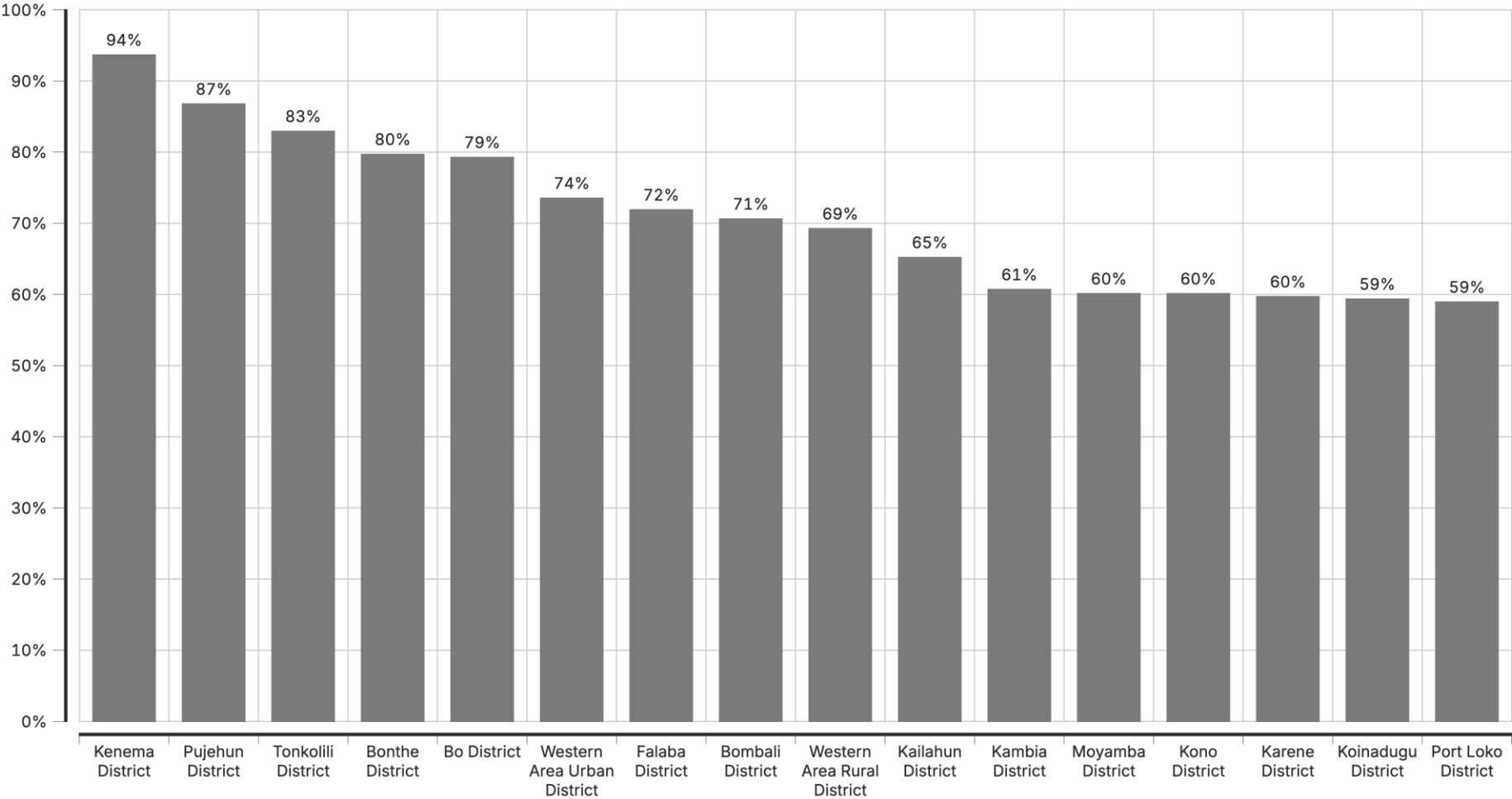
National



Coverage (subnational)

Sub-national level coverage estimates, Antenatal client 4th visit

2025



Estimating service coverage from administrative data can provide more timely information on coverage trends, or highlight data quality concerns. Numerators are the volumes reported in HMIS, adjusted for data quality.

Service coverage comparison to DHIS2

- Service coverage is commonly produced in DHIS2
- Main differences between DHIS2 and FASTR approach
 - FASTR approach adjusts numerators for outliers and/or completeness
 - FASTR approach calculates denominators from a variety of methods and chooses the denominator which produces the most plausible coverage estimates (closest to survey estimates)
 - FASTR approach allows for visualization of survey coverage estimates + HMIS coverage estimates + national projections made by applying HMIS trends to the most recent survey data

Coverage module: Configuration parameters

Parameter	Description
Count value to use	Which adjusted count to use for coverage calculation
Level to calculate coverage for	Geographic levels for coverage estimation: national, provincial (admin area 2), or district (admin area 3)
Pregnancy loss rate	Proportion of pregnancies ending in loss before delivery
Twin rate	Proportion of deliveries resulting in twins
Stillbirth rate	Proportion of births that are stillbirths
Neonatal mortality rate	Deaths in first 28 days per live birth
Postneonatal mortality rate	Deaths from 28 days to 1 year per live birth
Infant mortality rate	Deaths before age 1 per live birth
Under 5 mortality rate	Deaths before age 5 per live birth

Country-specific mortality rates may be obtained from DHS reports, UN IGME, or national vital statistics.

Session 6: Creating a project

Activity: Creating a Project

- Set up a new project
- Configure project settings
- Select indicators and time periods
- Apply best practices for project organization

Participants will create their first project

- Note since this is an admin feature all participants will NOT be doing this step. Instead, you will select one person to have admin rights, and they will help us walk through these steps.

Session 7: Creating visualizations

Activity: Creating Visualizations

- Explore available chart types
- Create and customize visualizations
- Export charts for use in reports

Participants will build visualizations from their analysis

Lunch Break

90 minutes • Back at 14:00

Session 8: Creating reports

Activity: Creating Reports

- Use report templates
- Generate automated reports
- Customize report content and layout

Participants will create their first quarterly report draft

Key messages and wrap-up

REFLECTIONS FROM PARTICIPANTS



STRENGTHENING HEALTH SYSTEMS AND RMNCAH-N OUTCOMES THROUGH RAPID CYCLE ANALYTICS AND DATA USE

*Country Workshop: Introduction to FASTR
RMNCAH-N service Use Monitoring*

Day 3



RECAP OF DAY 2

Agenda

Time	Agenda	Facilitator/Presenter
09:00–09:15	Overview of Day 3 agenda	GFF FASTR team
Session 8: Interpretation of visualizations		
09:15–10:15	Approaches to support interpretation	GFF FASTR team
Session 9: Creating a Q4 2025 report		
10:15–12:30	Creating short and long reports with country context	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 9 (cont'd): Creating a Q4 2025 report		
14:00–15:00	Continue report creation with country context	GFF FASTR team
Session 10: Presenting reports		
14:30–15:30	Present reports, group feedback	GFF FASTR team

Session 9: Interpretation of visualizations

Completeness: Percent of monthly values that are complete

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

For a given indicator in a given time period, the percent of monthly values that are complete:

% complete = # monthly values that are complete / total N of monthly values

90% or above

80% to 89%

Below 80%

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

Analytical thinking & interpretation

Interpretation connects **data patterns** to **programmatic meaning**.

For every FASTR output, ask three questions:

- **What does it show?** — Describe the pattern accurately
- **Why might that be?** — Consider multiple explanations
- **What should we do?** — Identify next steps or actions

Moving from numbers to insights requires context, critical thinking, and programmatic knowledge.

Critical thinking checklist

Before drawing conclusions from FASTR outputs, ask:

- **Data quality or real change?** Could this pattern reflect reporting issues rather than service delivery changes?
- **What external factors apply?** Policy changes, seasonal patterns, shocks, or campaigns?
- **Does it make programmatic sense?** Would this pattern be expected given what we know about the health system?
- **What does disaggregation reveal?** National trends can mask important regional variation
- **What additional information is needed?** What context would strengthen or challenge this interpretation?

Common interpretation pitfalls

Aggregation masking

National averages can hide critical regional variation. Always examine subnational patterns.

Completeness $\neq$ representativeness

High reporting completeness doesn't mean data captures all service delivery (private sector, informal care).

Missing context

Patterns without context are meaningless. Consider seasonality, policy changes, and external shocks.

Over-interpreting noise

Short-term fluctuations may be normal variation. Focus on sustained trends and significant deviations.

Interpreting FASTR module outputs

Data quality assessment

Low DQA scores may reflect data entry practices rather than service quality. Focus on identifying the specific dimension (completeness, outliers, consistency) driving issues.

Service utilization

Distinguish seasonal patterns from real disruptions. Year-over-year comparisons help isolate programmatic changes from normal fluctuation.

Coverage estimates

Coverage >100% signals reconciliation needed between HMIS and survey data. Consider whether the denominator source matches the service delivery context.

Across all modules

Examine patterns at multiple geographic levels — subnational disaggregation often reveals the true story.

Session 10: Creating a Q4 2025 report

Generating quarterly reporting products

Content to be developed

This section will cover:

- Quarterly reporting workflow
- Using the FASTR platform for automated reports
- Quality assurance for reports
- Distribution and feedback mechanisms

Lunch Break

90 minutes • Back at 14:00

Session 11: Presenting reports

Presenting reports and group feedback

Content to be developed

Day 4

Recap

On Day 3, we covered:

- Interpretation of FASTR visualizations
- Creation of the Q4 2025 quarterly report
- Presentation of findings and group feedback
- Action planning for continued platform use

Day 4 Focus

Today we design the Health Facility Assessment:

- Overview of the FASTR HFA phone survey
- Review questionnaire structure
- Adapt questionnaire for Zambia context
- Plan HFA priorities and data use

Key messages and wrap-up

REFLECTIONS FROM PARTICIPANTS



STRENGTHENING HEALTH SYSTEMS AND RMNCAH-N OUTCOMES THROUGH RAPID CYCLE ANALYTICS AND DATA USE

*Country Workshop: Introduction to FASTR
RMNCAH-N service Use Monitoring*

Day 4



RECAP OF DAY 3

Session 12: Overview of FASTR HFA phone survey

Agenda

Time	Agenda	Facilitator/Presenter
09:00–09:15	Overview of Day 4 agenda	GFF FASTR team
Session 12: Overview of FASTR HFA phone survey		
09:15–10:15	HFA overview and questionnaire adaptation guidelines	GFF FASTR team
Session 13: Questionnaire adaptation to the Zambian context		
10:15–12:30	Review questionnaire + hands-on adaptation	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 14: Questionnaire adaptation (cont'd)		
14:00–15:00	Continue questionnaire adaptation (in groups)	GFF FASTR team
Session 15: Discussion: HFA adapted questionnaire		
14:30–15:30	Discuss adapted questionnaire in plenary	GFF FASTR team
Session 16: Discussion: HFA priorities and data use		
15:30–16:30	HFA priorities and data use case in Zambia	GFF FASTR team
Session 17: Action planning and wrap-up		
16:30–17:00	Key messages and wrap-up	GFF FASTR team

Health Facility Assessment Phone Survey

Placeholder. Content to be developed

Lunch Break

90 minutes • Back at 14:00

Next Steps & Action Planning

Key actions to take after this working session:

- Finalize adapted phone survey questionnaire
- Complete first quarterly report draft
- Train remaining MoH team on data downloader tools
- Prepare roadmap for participation in Abuja workshop

Quarterly Reporting Cycle

Establish a sustainable reporting rhythm:

- Define quarterly report timeline and responsibilities
- Set up data extraction schedule
- Establish review and approval process
- Plan dissemination to stakeholders

Continued Platform Usage

Maintain momentum with the FASTR platform:

- Regular data updates and quality checks
- Expanding analysis to additional indicators
- Building capacity across MoH teams
- Documentation of lessons learned

Workshop Closing

Thank you for your participation!

Over the past four days, we have:

- Configured the FASTR platform for Zambia
- Created the first quarterly RMNCAH-N report
- Adapted the HFA questionnaire for local context
- Built capacity for ongoing analysis and reporting

Stay Connected

Continue the journey:

- Platform access and support
- Quarterly report submissions
- Multi-country workshop in Abuja
- Ongoing technical assistance from GFF team

Thank You

FASTR Working Session – Zambia

January 27–30, 2026

Lusaka

Contact Information

FASTR Team

Email:

Website: <https://www.globalfinancingfacility.org/>



About FASTR | Access the FASTR resource repository

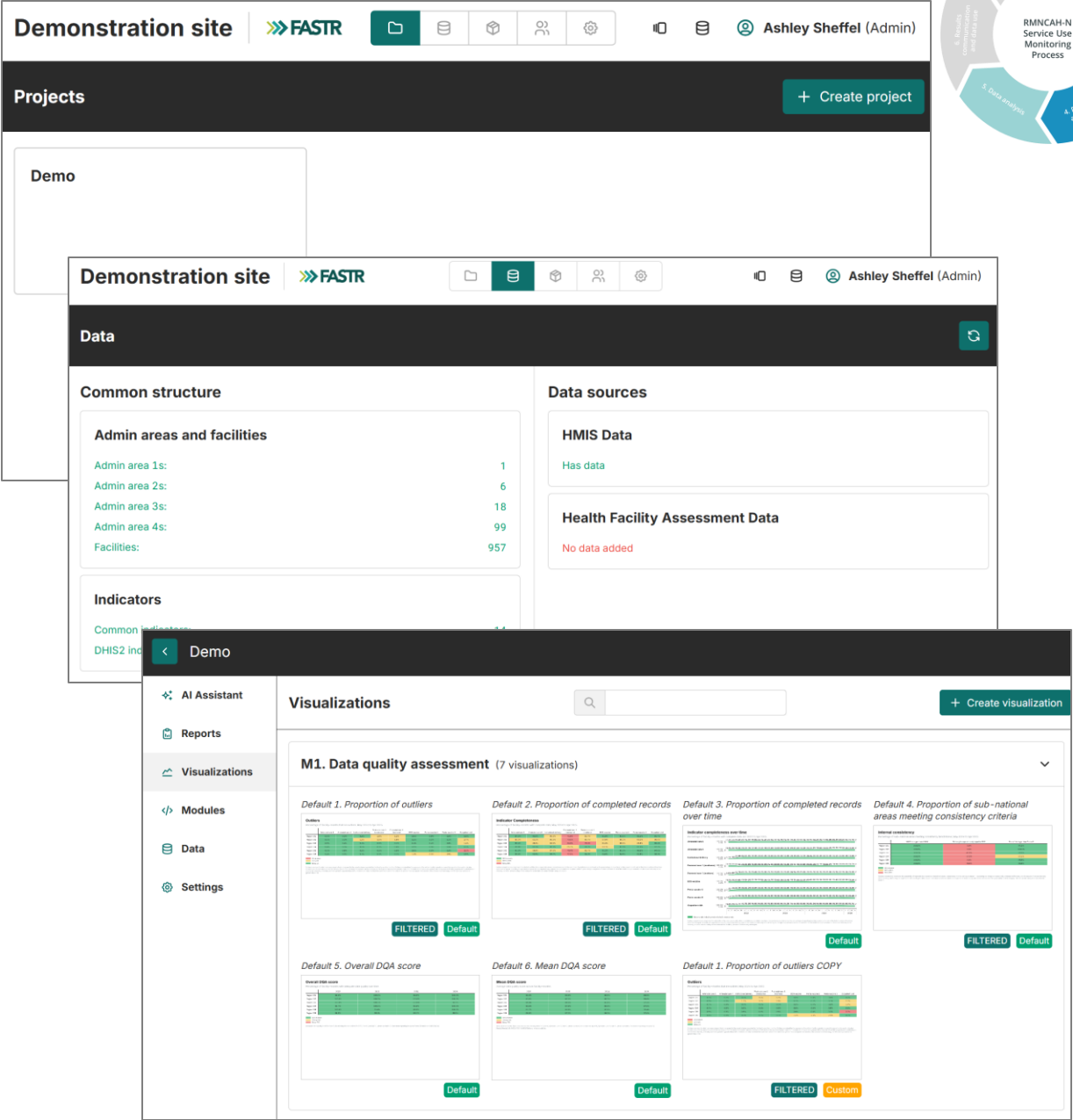
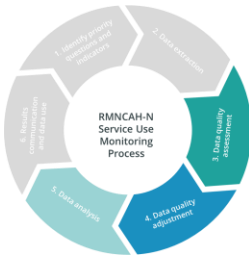
Contact us at fastr@worldbank.org

Data quality assessment, adjustment, analysis

The FASTR analytics platform is a web-based tool designed to support data quality assessment, adjustment, and analysis for routine health data.

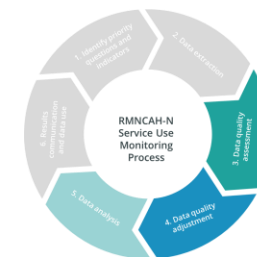
It allows users to upload and analyze data from various sources, including DHIS2, with built-in statistical methods to generate an adjusted dataset and run priority analyses on selected indicators.

The platform provides a user-friendly interface for running analyses and offers flexible options for visualizing and exporting results.



What's comes next?

- The upcoming workshop will focus on how to use the FASTR analytics platform to update your disruptions analysis and create additional outputs to serve country-specific use cases.
- You will receive an email on how to register and access your FASTR analytics platform. Please be sure you can login to the platform **BEFORE** arriving at the workshop.



Tracking Disruptions in Essential Services Using HMIS Data in

Methodology: Service Utilization Assessment

Purpose:

Track changes in health service use over time, identifying where services fall below or rise above expected patterns.

How it works:

- Uses routine HMIS data, cleaned for outliers and missing values.
- Builds an "expected" trend line for each service, adjusting for seasonality and historical trends in service utilization.
- Compares actual service volumes to expected levels.

Measuring impact:

- Flagged disruption periods are analyzed to estimate how much service volumes changed compared to what was expected.

Results are shown as:

How to interpret:

- Red shaded areas =
- Green shaded areas =
- These are signals, not

More details on the methodology can be found at <https://github.com/WHO-HQ-IMR/FASTR>

PNC visits rise above expected levels in 2025

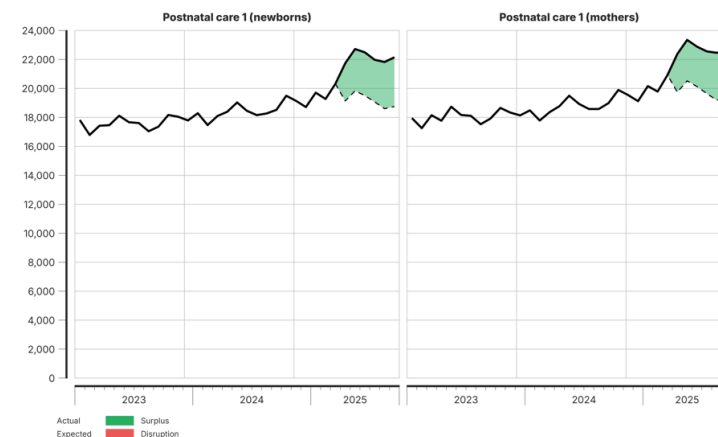
Newborn PNC visits rose above expected levels in 2025, reaching ~22,700–22,900 by May–June, which corresponds to surpluses of ~14–15%.

Maternal PNC visits showed a very similar pattern, rising to ~23,300 in May and ~22,900 in June, equivalent to surpluses of ~14% above expected levels.

Across 2025, PNC for newborns consistently reached ~97–99% of maternal PNC volumes, showing strong alignment between mother and newborn follow-up. However, both remained slightly below delivery volumes, indicating modest gaps in completing timely postpartum care for all mothers and babies.

Comparing reported service use to expected trends, nationally

Jan 2023 to Sep 2025



This graph quantifies changes in service volume compared to historical trends and accounting for seasonality. These signals should be triangulated to other data and contextual knowledge to determine if the results are an artifact of data quality. Unexpected volume changes are estimated by comparing the observed volume to the expected volume based on historical trends and seasonality. Previous large unexpected changes in the historical data are removed. This analysis is an interrupted time series regression with facility-level fixed effects.